1

(i) Show that the equation $\frac{\tan\theta}{\cos\theta} = 1$ may be rewritten as $\sin\theta = 1 - \sin^2\theta$

[2]

(ii) Hence solve the equation $\frac{\tan\theta}{\cos\theta} = 1$ for $0° \leqslant \theta \leqslant 360°$.

[3]

2 Fig. 7 shows a sketch of a village green ABC which is bounded by three straight roads. AB = 92 m, BC = 75 m and AC = 105 m.

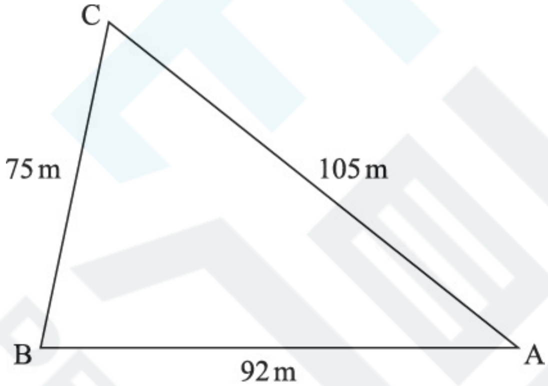


Fig. 7

Calculate the area of the village green.

[5]

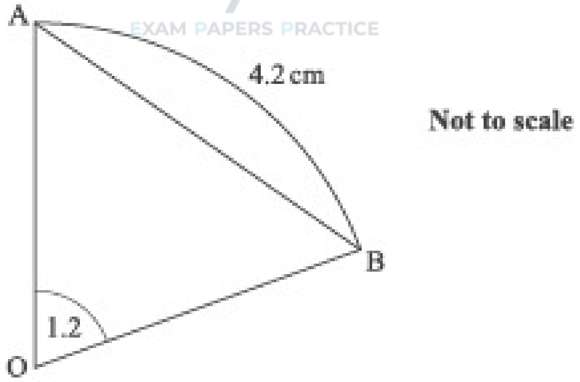


Fig. 4

Fig. 4 shows sector OAB with sector angle 1.2 radians and arc length 4.2 cm. It also shows chord AB.

(i) Find the radius of this sector.

[2]

(ii) Calculate the perpendicular distance of the chord AB from O.

[2]

4 Fig. 10.1 shows Jean's back garden. This is a quadrilateral ABCD with dimensions as shown.

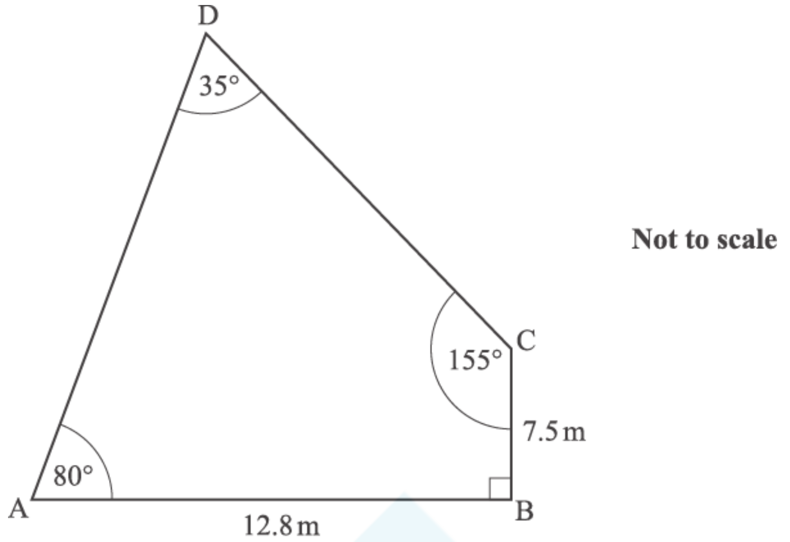


Fig. 10.1

(i)

A Calculate AC and angle ACB. Hence calculate AD. [6]

B Calculate the area of the garden. [3]

(ii) The shape of the fence panels used in the garden is shown in Fig. 10.2. EH is the arc of a sector of a circle with centre at the midpoint, M, of side FG, and sector angle 1.1 radians, as shown. FG = 1.8 m.

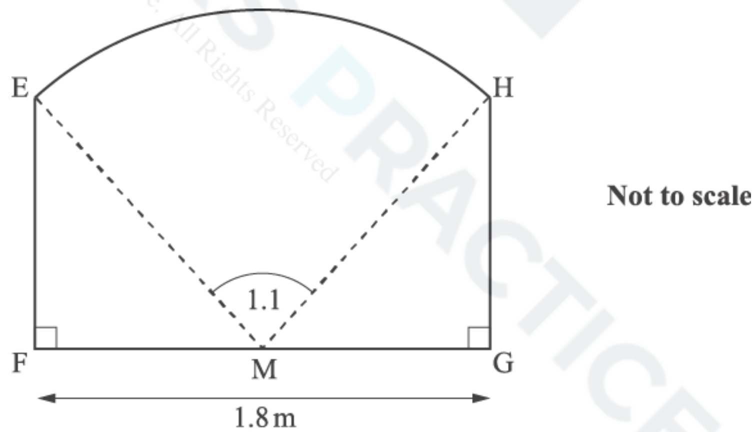


Fig. 10.2

Calculate the area of one of these fence panels. [5]

5

(i) Starting with an equilateral triangle, prove that $\cos 30° = \frac{\sqrt{3}}{2}$. [2]

(ii) Solve the equation $2 \sin\theta = -1$ for $0 \leqslant \theta \leqslant 2\pi$, giving your answers in terms of π. [3]

6 Solve the equation $\tan 2\theta = 3$ for $0° < \theta < 360°$.

[3]

7 Simplify $\frac{\sqrt{1-\cos^2\theta}}{\tan\theta}$, where θ is an acute angle.

[3]

8

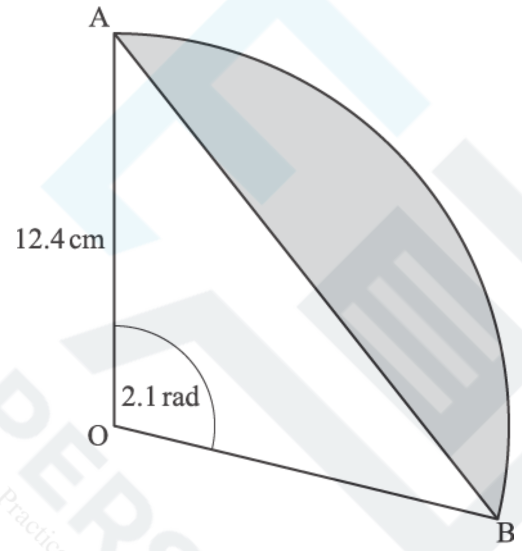


Fig. 6

A circle with centre O has radius 12.4 cm. A segment of the circle is shown shaded in Fig. 6. The segment is bounded by the arc AB and the chord AB, where the angle AOB is 2.1 radians. Calculate the area of the segment.

[4]

9

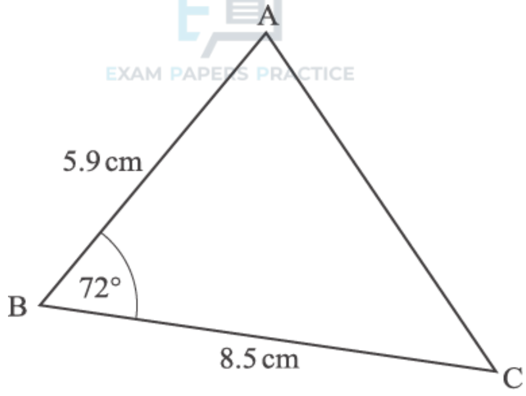


Fig. 5

Fig. 5 shows triangle ABC, where angle ABC = 72°, AB = 5.9 cm and BC = 8.5 cm. Calculate the length of AC.

[3]

10

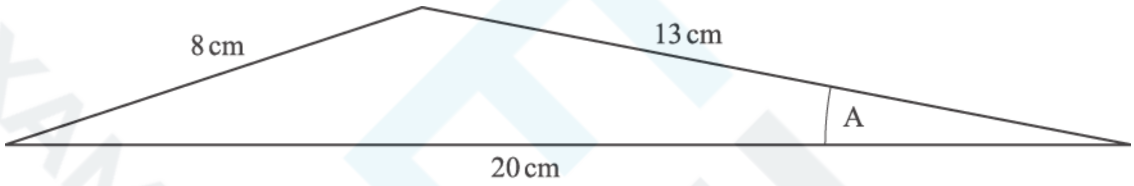


Fig. 9.1

(i) Jean is designing a model aeroplane. Fig. 9.1 shows her first sketch of the wing's cross-section. Calculate angle A and the area of the cross-section.

[5]

(ii) Jean then modifies her design for the wing. Fig. 9.2 shows the new cross-section, with 1 unit for each of *x* and *y* representing one centimetre.

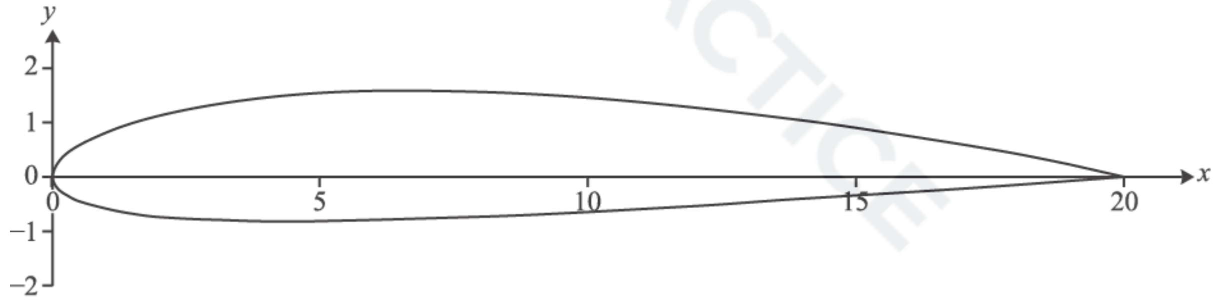


Fig. 9.2

Here are some of the coordinates that Jean used to draw the new cross-section.

Upper surface		Lower surface	
x	y	x	y
0	0	0	0
4	1.45	4	–0.85
8	1.56	8	–0.76
12	1.27	12	–0.55
16	1.04	16	–0.30
20	0	20	0

Use the trapezium rule with trapezia of width 4 cm to calculate an estimate of the area of this cross–section.

[6]

11 Show that the equation $\sin^2 x = 3\cos x - 2$ can be expressed as a quadratic equation in $\cos x$ and hence solve the equation for values of 3^x between 0 and 2π.

[5]

12 A sector of a circle has angle 1.5 radians and area 27 cm^2. Find the perimeter of the sector.

[4]

13

(i) Show that, when x is an acute angle, $\tan x\sqrt{1-\sin^2 x} = \sin x$.

[2]

(ii) Solve $4\sin^2 y = \sin y$ for $0° \leqslant y \leqslant 360°$.

[3]

14 A sector of a circle has radius r cm and sector angle θ radians. It is divided into two regions, A and B. Region A is an isosceles triangle with the equal sides being of length a cm, as shown in Fig. 6.

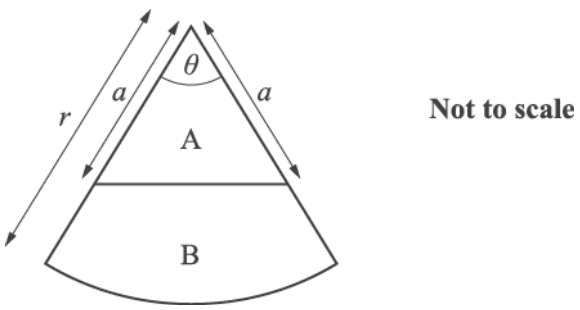


Fig. 6

(i) Express the area of B in terms of a, r and θ.

[2]

(ii) Given that $r = 12$ and $\theta = 0.8$, find the value of a for which the areas of A and B are equal. Give your answer correct to 3 significant figures.

[2]

15

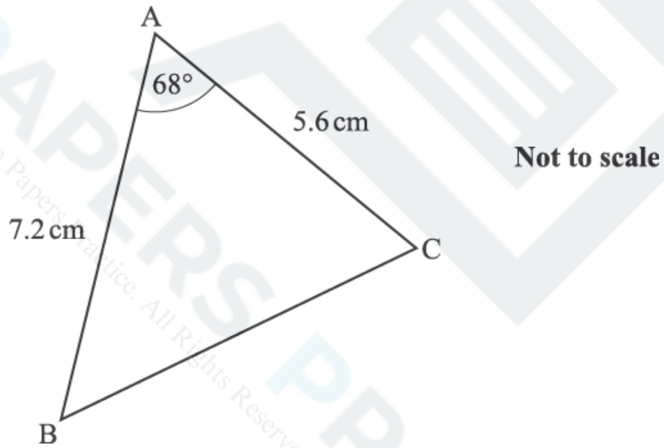


Fig. 4

Fig. 4 shows triangle ABC, where AB = 7.2 cm, AC = 5.6 cm and angle BAC = 68°.
Calculate the size of angle ACB.

[5]

16 Fig. 4 shows the curve $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = a + \cos bx,\ 0 \leqslant x \leqslant 2\pi,$$

and a and b are positive constants. The curve has stationary points at (0, 3) and (2π, 1).

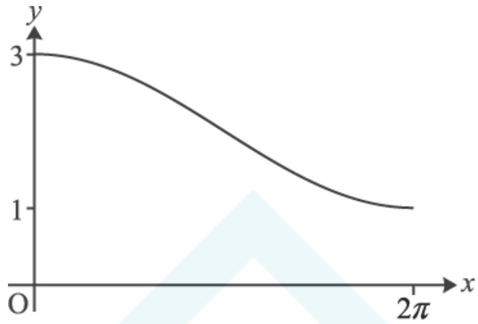


Fig. 4

(i) Find a and b.

[2]

(ii) Find $\mathrm{f}^{-1}(x)$, and state its domain and range.

[5]

17 Solve each of the following equations, giving your answers in exact form.

(i) $6 \arcsin x - \pi = 0$.

[2]

(ii) $\arcsin x = \arccos x$.

[2]

18 In Fig. 6, ABC, ACD and AED are right-angled triangles and BC = 1 unit. Angles CAB and CAD are θ and ϕ respectively.

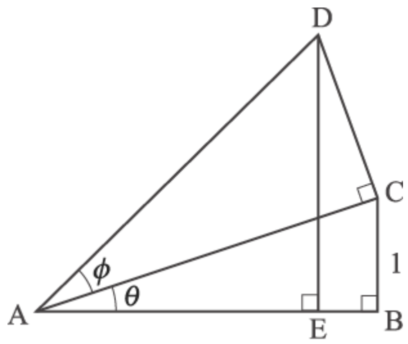


Fig. 6

(i) Find AC and AD in terms of θ and ϕ.

[2]

(ii) Hence show that $\mathrm{DE} = 1 + \dfrac{\tan\phi}{\tan\theta}$

[3]

19 Solve the equation 2 $\sec^2\theta = 5\tan\theta$, for $0 \le \theta \le \pi$.

[6]

20 The motion of a particle is modelled by the differential equation

$$v\frac{\mathrm{d}v}{\mathrm{d}x} + 4x = 0,$$

where x is its displacement from a fixed point, and v is its velocity.

Initially $x = 1$ and $v = 4$.

(i) Solve the differential equation to show that $v^2 = 20 - 4x^2$.

[4]

Now consider motion for which $x = \cos 2t + 2 \sin 2t$, where x is the displacement from a fixed point at time t.

(ii) Verify that, when $t = 0$, $x = 1$. Use the fact that $v = \frac{\mathrm{d}x}{\mathrm{d}t}$ to verify that when $t = 0$, $v = 4$.

[4]

(iii) Express x in the form $R \cos(2t - \alpha)$, where R and α are constants to be determined, and obtain the corresponding expression for v. Hence or otherwise verify that, for this motion too, $v^2 = 20 - 4x^2$.

[7]

(iv) Use your answers to part **(iii)** to find the maximum value of x, and the earliest time at which x reaches this maximum value.

[3]

21

Using appropriate right-angled triangles, show that $\tan 45° = 1$ and $\tan 30° = \frac{1}{\sqrt{3}}$.

Hence show that $\tan 75° = 2 + \sqrt{3}$.

[7]

22 Show that the equation $\operatorname{cosec} x + 5 \cot x = 3 \sin x$ may be rearranged as

$$3 \cos^2 x + 5 \cos x - 2 = 0.$$

Hence solve the equation for $0° \leqslant x \leqslant 360°$, giving your answers to 1 decimal place.

[7]

23

(i) Show that $\cos(\alpha+\beta) = \dfrac{1-\tan\alpha\tan\beta}{\sec\alpha\sec\beta}$.

[3]

(ii) Hence show that $\cos 2\alpha = \dfrac{1-\tan^2\alpha}{1+\tan^2\alpha}$.

[2]

(iii) Hence or otherwise solve the equation $\dfrac{1-\tan^2\theta}{1+\tan^2\theta} = \dfrac{1}{2}$ for $0° \le \theta \le 180°$.

[3]

24 You are given that $f(x) = \cos x + \lambda \sin x$ where λ is a positive constant.

(i) Express $f(x)$ in the form $R\cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < \frac{1}{2}\pi$, giving R and α in terms of λ.

[4]

(ii) Given that the maximum value (as x varies) of $f(x)$ is 2, find R, λ and α, giving your answers in exact form.

[4]

25 Express $6\cos 2\theta + \sin\theta$ in terms of $\sin\theta$.

Hence solve the equation $6\cos 2\theta + \sin\theta = 0$, for $0° \le \theta \le 360°$.

[7]

26 P is a general point on the curve with parametric equations $x = 2t,\ y = \frac{2}{t}$. This is shown in Fig. 6. The tangent at P intersects the x - and y-axes at the points Q and R respectively.

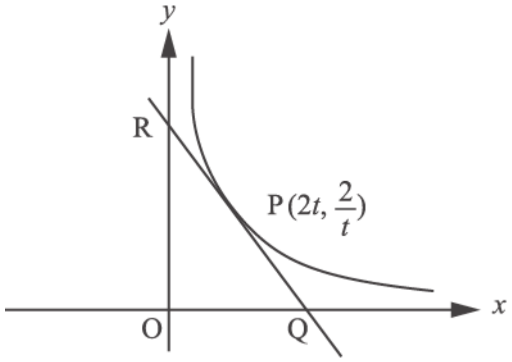


Fig. 6

Show that the area of the triangle OQR, where O is the origin, is independent of t.

[7]

27 In Fig. 5, triangles ABC, ACD and ADE are all right-angled, and angles BAC, CAD and DAE are all θ. AB = x and AE = $2x$.

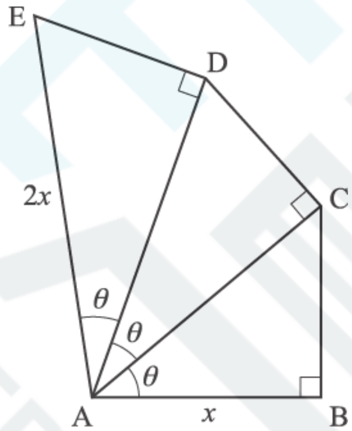


Fig. 5

(i) Show that $\sec^3\theta = 2$.

[3]

(ii) Hence show the ratio of lengths ED to CB is $2^{\frac{2}{3}} : 1$.

[4]

28 Solve the equation $2 \sin 2\theta = 1 + \cos 2\theta$ for $0° \leqslant \theta \leqslant 180°$.

[5]

29 Express $\cos \theta - 3 \sin \theta$ in the form $R \cos(\theta + \alpha)$, where $R > 0$ and $0 < \alpha < \frac{\pi}{2}$.

Hence show that the equation $\cos \theta - 3 \sin \theta = 4$ has no solution.

[6]

(i) Use Equation (3) to calculate the declination of the sun on February 2nd.

[3]

(ii) The town of Boston, in Lincolnshire, has latitude 53° north and longitude 0°.

Calculate the time of sunset in Boston on February 2nd.

Give your answer in hours and minutes using the 24-hour clock.

[4]

31 In lines 94 and 95 the article says

"Fig. 8 shows you that at latitude 60° north the terminator passes approximately through time +9 hours and ?9 hours so that there are about 18 hours of daylight."

Use Equation (4) to check the accuracy of the figure of 18 hours.

[4]

32 The graph is a copy of Fig. 6.

The article says that it shows the terminator in the cases where the sun has declination 10° north, 1° north,5° south and 15° south.

Identify which curve (A, B, C or D) relates to which declination.

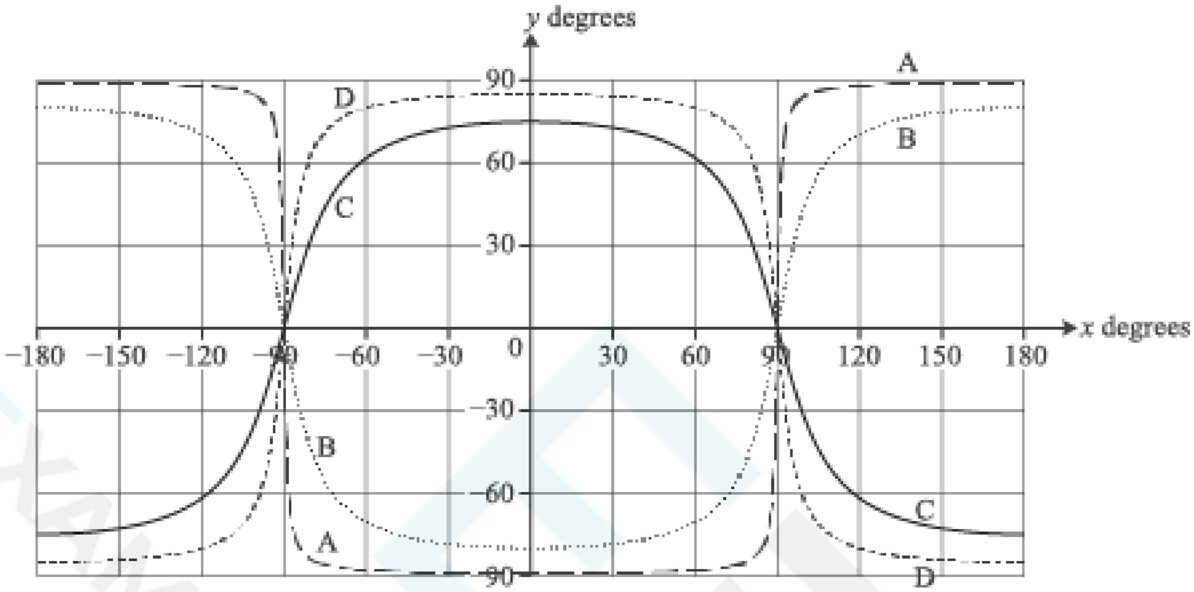


Fig. 6

10° north:

1° north:

5° south:

15° south:

---

[2]

33 Use Fig. 8 to estimate the difference in the length of daylight between places with latitudes of 30° south and 60° south on the day for which the graph applies.

[3]

34 The diagram is a copy of Fig. 4.

R is a place with latitude 45° north and longitude 60° west. Show the position of R on the diagram.

M is the sub-solar point. It is on the Greenwich meridian and the declination of the sun is $+20°$. Show the position of M on the diagram.

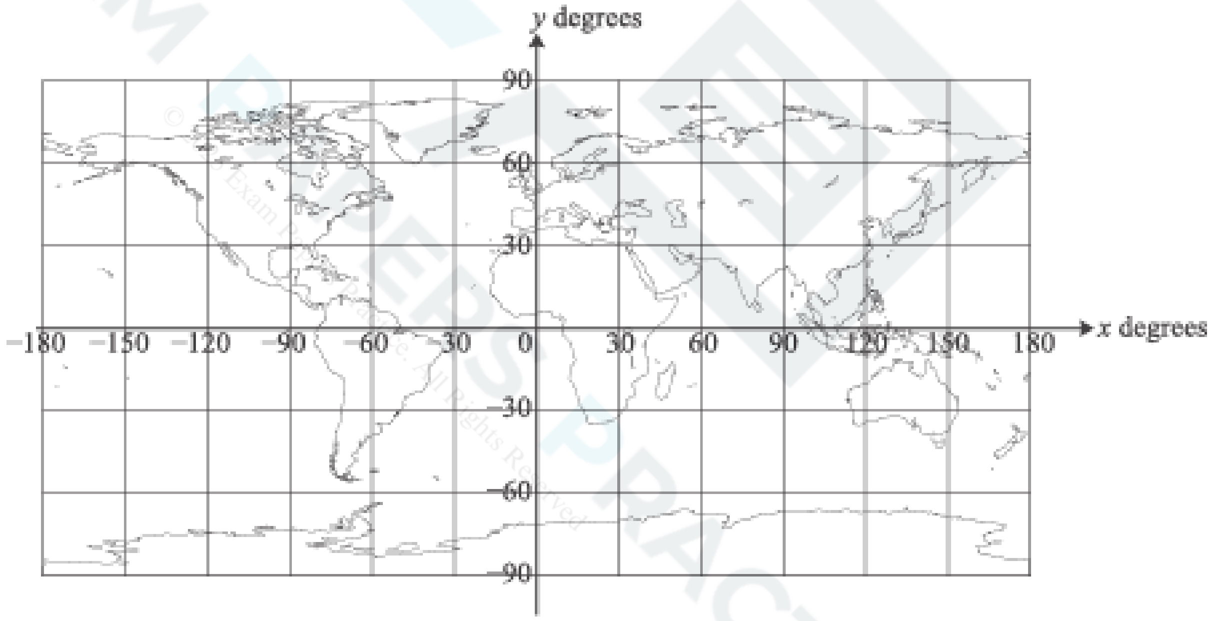


Fig. 4

[2]

35 Evaluate $\int_0^{\frac{\pi}{12}} \cos 3x \, dx$, giving your answer in exact form. [3]

36 A triangular field has sides of length 100 m, 120 m and 135 m.

(a) Find the area of the field. **[5]**

(b) Explain why it would not be reasonable to expect your answer in **(a)** to be accurate to the nearest square metre. **[1]**

37 (a) The graph of $y = 3\sin^2\theta$ for $0° \le \theta \le 360°$ is shown in Fig. 6. On Fig. 6, sketch the graph of $y = 2\cos\theta$ for $0° \le \theta \le 360°$. [2]

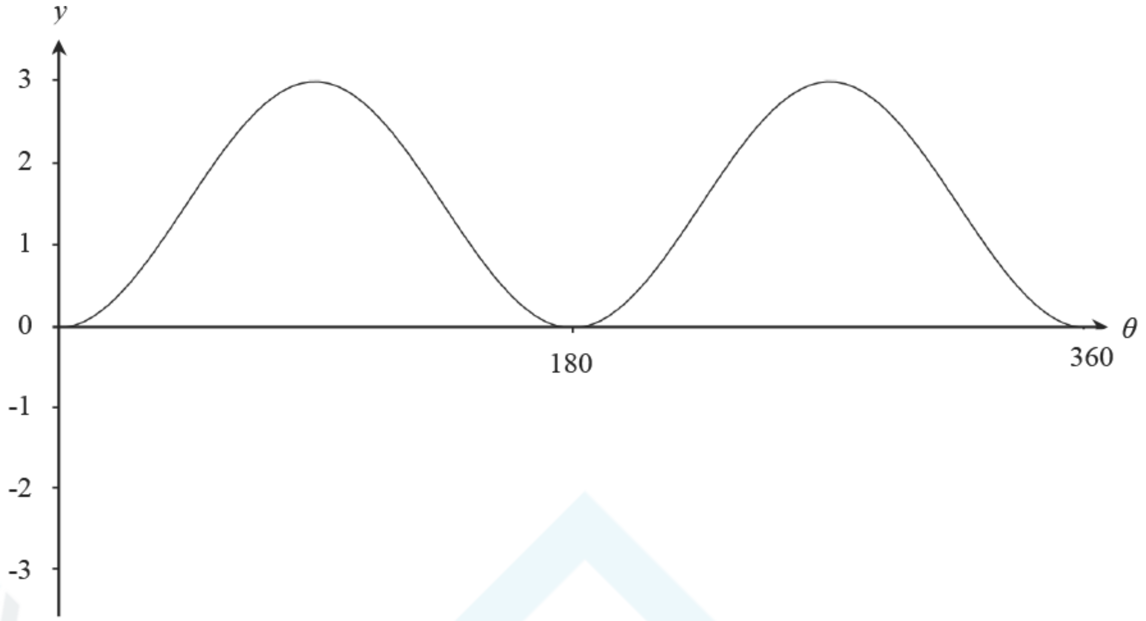


Fig. 6

(b) In this question you must show detailed reasoning.

Determine the values of θ , $0° \leq \theta \leq 360°$, for which the two graphs cross. **[6]**

38 Given that arcsin x = arccos y, prove that $x^2 + y^2 = 1$. [Hint: Let arcsin $x = \theta$] **[3]**

39 Fig. 1 shows a sector of a circle of radius 7 cm. The area of the sector is 5 cm^2.

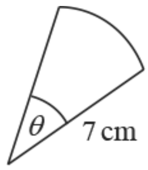


Fig. 1

[2]

Find the angle θ in radians.

40 A woman is pulling a loaded sledge along horizontal ground. The only resistance to motion of the sledge is due to friction between it and the ground.

Fig. 5

At first she pulls with a force of 100N inclined at 32° to the horizontal, as shown in Fig.5, but the sledge does not move.

(a) Determine the frictional force between the ground and the sledge.
Give your answer correct to 3 significant figures. **[2]**

(b) Next she pulls with a force of 100N inclined at a smaller angle to the horizontal. The sledge still does not move.

[2]

Compare the frictional force in this new situation with that in part **(a)**, justifying your answer.

41 **In this question you must show detailed reasoning.**

Fig. 5 shows the circle with equation $(x-4)^2 + (y-1)^2 = 10$. The points (1, 0) and (7, 0) lie on the circle. The point C is the centre of the circle.

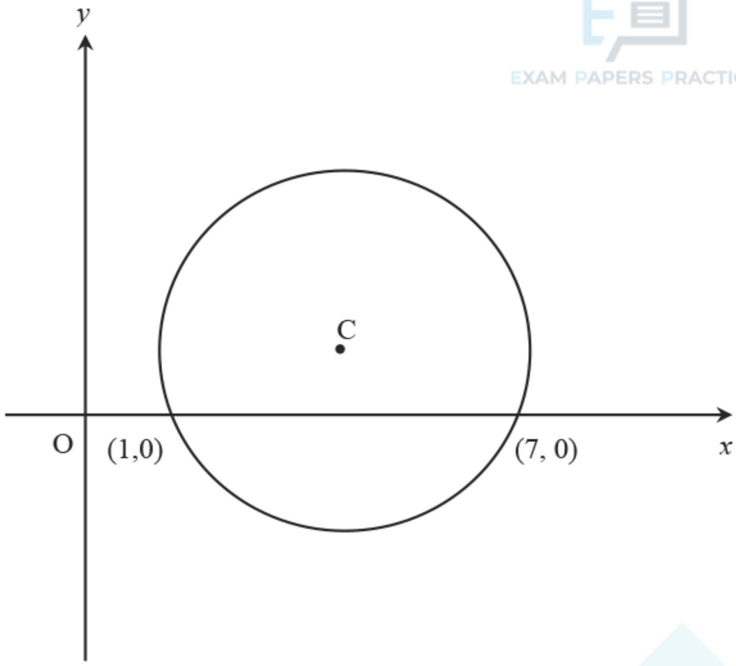


Fig. 5

Find the area of the part of the circle below the *x*-axis. **[5]**

42 In Fig. 8, OAB is a thin bent rod, with OA = 1 m, AB = 2 m and angle OAB = 120°. Angles θ , φ and h are as shown in Fig. 8.

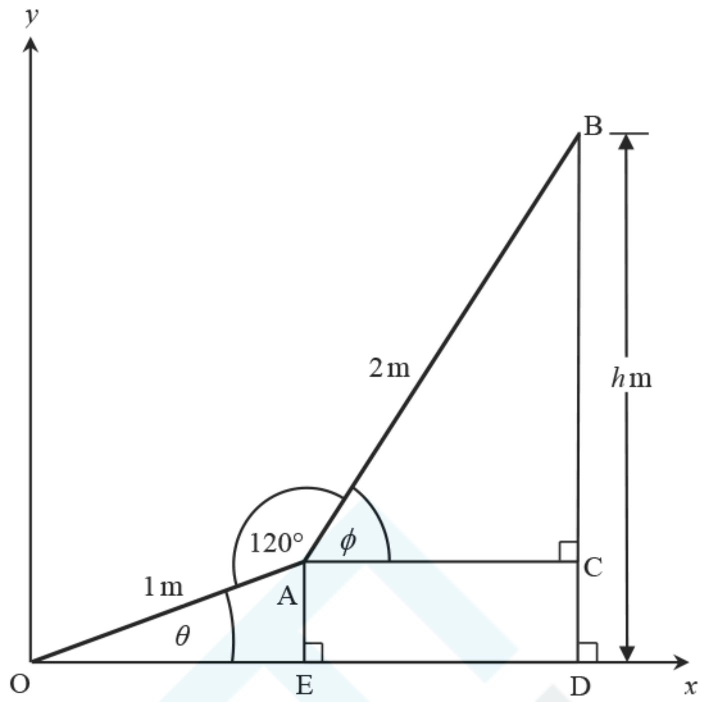


Fig. 8

(a) Show that $h = \sin\theta + 2\sin(\theta + 60°)$. **[3]**

The rod is free to rotate about the origin so that θ and φ vary. You may assume that the result for h in part **(a)** holds for all values of θ.

(b) Find an angle θ for which $h = 0$. **[5]**

43

(a) Express $\cos\theta + 2\sin\theta$ in the form $R\cos(\theta - a)$, where $0 < \alpha < \frac{1}{2}\pi$ and R is positive and given in exact form. **[4]**

The function $f(\theta)$ is defined by $f(\theta) = \frac{1}{(k + \cos\theta + 2\sin\theta)}$, $0 \leq \theta \leq 2\pi$, k is a constant.

(b) The maximum value of $f(\theta)$ is $\frac{(3+\sqrt{5})}{4}$. Find the value of k. **[3]**

44 The curve $y = f(x)$ is defined by the function $f(x) = e^{-x} \sin x$ with domain $0 \le x \le 4\pi$.

(a) (i) Show that the x-coordinates of the stationary points of the curve $y = f(x)$, when arranged in increasing order, form an arithmetic sequence.

(ii) Show that the corresponding y-coordinates form a geometric sequence. **[9]**

(b) Would the result still hold with a larger domain? Give reasons for your answer **[1]**

45 (See Insert for Specimen 64003.) Explain why the smaller regular hexagon in Fig. C1 has perimeter 6. **[1]**

46 (See Insert for Specimen 64003.) Show that the larger regular hexagon in Fig. C1 has perimeter $4\sqrt{3}$. **[3]**

47 (See Insert for Specimen 64003.) Fig. 15 shows a unit circle and the escribed regular polygon with 12 edges.

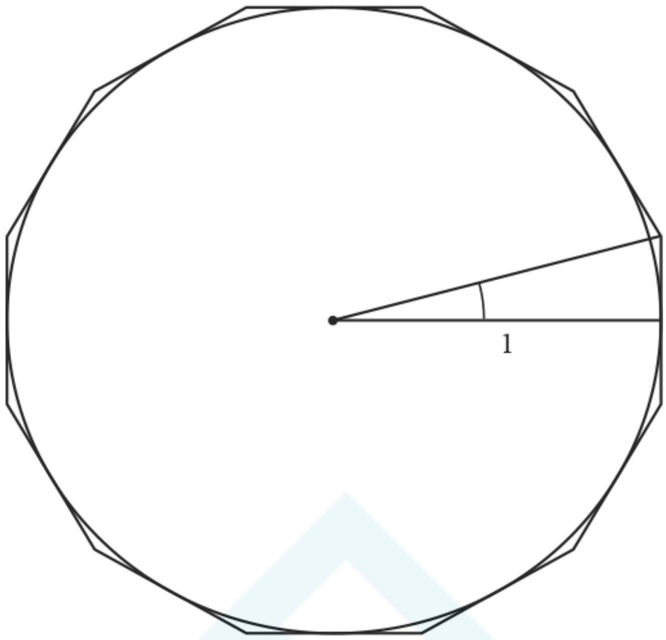


Fig. 15

(a) Show that the perimeter of the polygon is 24 tan15° . [2]

(b) Using the formula for tan $(\theta - \phi)$ show that the perimeter of the polygon is $48 - 24\sqrt{3}$. [3]

48 **In this question you must show detailed reasoning.**

Solve the equation

$$2\tan\theta + \cos\theta = 0$$

in the range $0° < \theta < 360°$. **[7]**

49 Solve the equation $\cos 2\theta = 0.3$ for $0° \leq \theta < 360°$. **[3]**

50 A car travels along a straight track for 5 seconds. Its displacement s metres after t seconds is given by

$$s = 3t + 0.1t^3 .$$

Show that the car does not have constant acceleration. **[3]**

51 Fig. 6.1 shows a shape consisting of an isosceles triangle ABC and a semicircle with AC as diameter. Angle ABC $= \frac{1}{6}\pi$ radians and AB = BC = 10 cm.

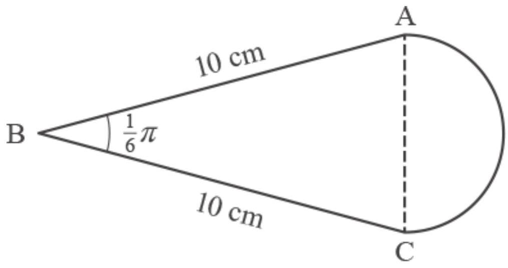


Fig. 6.1

(a) Calculate the total area of the shape. **[4]**

Fig. 6.2 shows a sector DEF of a circle of radius 10 cm. Angle DEF = θ radians.

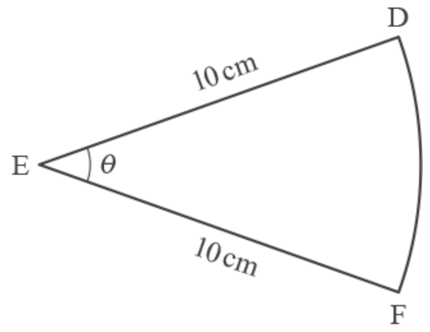


Fig. 6.2

(b) The shapes in Fig. 6.1 and Fig. 6.2 have the same area. Find θ, giving your answer correct to 3 significant figures. **[2]**

52 A globe hangs on the end of a light wire that makes an angle of θ with the vertical. It is held in place by a light horizontal string, as shown in Fig. 7.

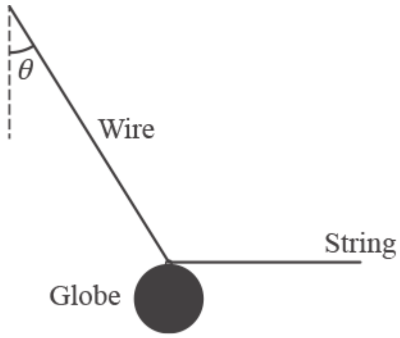


Fig. 7

Find the set of values of θ for which the tension in the string is less than the weight of the globe. **[4]**

53 **(a)** Prove the identity $\sec\theta - \cos\theta \equiv \tan\theta \sin\theta$. **[4]**

(b) Hence or otherwise solve the equation $\sec\theta - \cos\theta = \frac{1}{2}\tan\theta$ for $0 \leq \theta < 2\pi$. **[4]**

54 Gerry has been collecting data about the ladybird population in his locality, where they appear mainly during the spring, summer and autumn. He has carried out surveys on many days over a three-month period in the summer, and has recorded the values of N, the number of ladybirds seen, and t, the time in days after 1 January. Using graph-drawing software, he has found that the part of the quadratic curve shown in Fig. 13.1 is a good fit for his data.

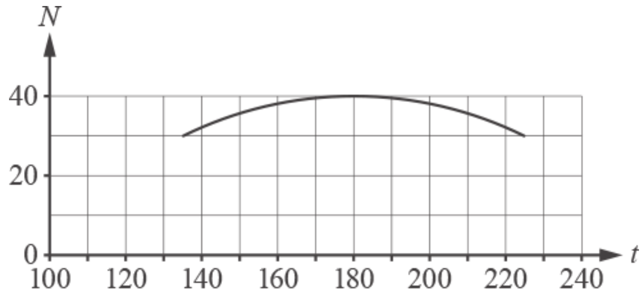


Fig. 13.1

This curve has a maximum point at (180, 40) and the point (135, 30) also lies on the curve. Gerry uses the equation of the curve in the form

$$N = a - b(t - c)^2$$

as a model for the ladybird population.

(a) Find the values of a and c and show that $b = \frac{2}{405}$. [3]

(b) (i) Find the number of ladybirds predicted by the model when $t = 250$. [1]

(ii) Explain why this predicted value may be unreliable. [1]

(c) (i) Explain why the model is not valid for $t < 90$. **[1]**

(ii) Give another range of values of t for which the model is not valid. **[1]**

With the software set to degrees, Gerry also produces the graph in Fig. 13.2 to model his data for suitable values of t. This graph is a transformation of $N = \cos t$; it passes through (90, 0) and has a maximum point at (180, 40).

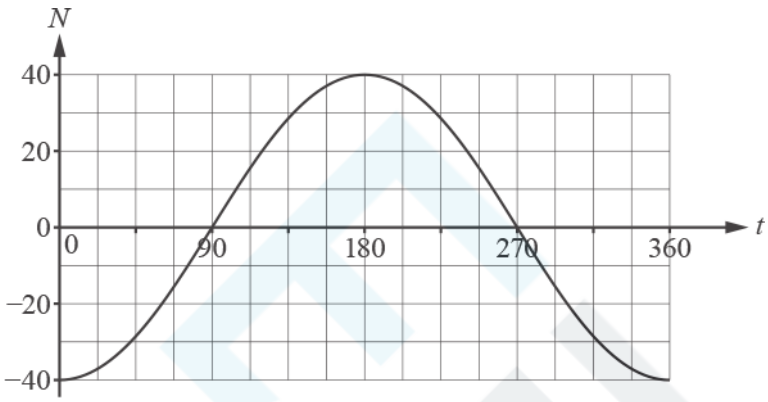


Fig. 13.2

(d) Describe a sequence of transformations that maps the curve $N = \cos t$ to the curve in Fig. 13.2. **[2]**

(e) Write down the equation for N in terms of t for this model. **[1]**

(f) (i) Explain how the model can be refined to have a period of 365 days. **[1]**

(ii) What effect, if any, does this refinement have on the maximum point of the graph? **[1]**

55

(a) Express $2 \cos \theta + 3 \sin \theta$ in the form $R \sin (\theta + \alpha)$, where $0 < \alpha < \frac{1}{2}\pi$ and R is a positive constant given in exact form. **[4]**

(b) Determine the set of values of k for which the curve $y = k + 2\cos x + 3\sin x$ lies completely above the x-axis. **[4]**

(c) Explain why the curve $y = \dfrac{1}{k + 2\cos x + 3\sin x}$ lies completely above the x-axis for the set of values of k found in part **(b)**. **[1]**

56 **(a)** Write down the exact values of tan 45° and tan 60°. **[1]**

(b) **In this question you must show detailed reasoning.**

Show that $\tan 15° = 2 - \sqrt{3}$. **[4]**

57 The sides of a triangle are of length 47, 53 and 94 units. Calculate the size of the largest angle. **[3]**

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Part marks and guidance	
1		i	$(\frac{\frac{\sin\theta}{\cos\theta}}{\cos\theta}) = 1$ oe	M1		
		i	sin θ = $\cos^2\theta$ and completion to given result	A1	www **Examiner's Comments** Many candidates answered this question well, although there were a number of attempted fudges using $\tan\theta = {}^{\cos\theta}/_{\sin\theta}$. Some adopted a scattergun approach and it was not always possible to follow their method.	
		ii	$\sin^2\theta + \sin\theta - 1 [= 0]$	M1	allow 1 on RHS if attempt to complete square	condone $y^2 + y - 1 = 0$
		ii	$[\sin\theta =] \frac{-1 \pm \sqrt{5}}{2}$ oe may be implied by correct answer	A1	may be implied by correct answers	mark to benefit of candidate ignore any work with negative root & condone omission of negative root with no comment eg M1 for 0.618…

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	[θ =] 38.17…, or 38.2 and 141.83…, 141.8 or 142	A1	Ignore extra values outside range, **A0** if extra values in range or in radians **NB** 0.6662 and 2.4754 if working in radian mode earns **M1A1A0** **Examiner's Comments** This defeated a significant minority of candidates. However, many obtained the correct quadratic equation. Most then went on to attempt factorisation, going wrong and failing to score. A minority successfully completed the square or used the formula. Many of these went on to score full marks, but some candidates missed the last mark because they presented extra values in the range, or because they didn't realise that further work was needed after obtaining the roots of the quadratic.	If unsupported, **B1** for one of these, **B2** for both. If both values correct with extra values in range, then **B1**. **NB** 0.6662 and 2.4754 to 3sf or more
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
2			$\cos A = \frac{105^2 + 92^2 - 75^2}{2 \times 105 \times 92}$ oe	M1	or $\cos B = \frac{75^2 + 92^2 - 105^2}{2 \times 75 \times 92}$ oe	or $\cos C = \frac{105^2 + 75^2 - 92^2}{2 \times 105 \times 75}$ oe
			0.717598…soi	A1	0.2220289…soi	0.519746…soi
			A = 44.14345…° soi [0.770448553…]	A1	B = 77.1717719……° soi [1.346901422]	C = 58.6847827…°soi [1.024242678 …] ignore minor errors due to premature rounding for second **A1** condone *A*, *B* or *C* wrongly attributed
			½ × 92 × 105 × sin (*their* A)	M1	or ½ × 75 × 92 × sin (*their* B)	or ½ × 75 × 105 × sin (*their* C)

Question			Answer/Indicative content	Marks	Part marks and guidance	
			3360 or 3361 to 3365	A1	**Examiner's Comments** Nearly all candidates adopted the expected approach successfully and achieved full marks. A few rounded the angle prematurely and lost the final mark. Some lost the last two marks by using “cos” instead of “sin” in the area formula, and similarly a very few candidates used “sin” instead of “cos” in the Cosine Rule. Most candidates went on to use the correct sides with the angle that had been found. After using the Cosine Rule successfully a few candidates opted for ½ base × height and about half of these did so successfully. A tiny minority of candidates used Hero's formula successfully. Only a small number treated the triangle as right angled and failed to score.	or M3 for $\sqrt{136(136-75)(136-105)(136-92)}$ A2 for correct answer 3360 or 3363 – 3364
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
3		i	$1.2r = 4.2$	M1	or $\frac{68.7549...}{360} \times 2\pi r = 4.2$ with θ to 3 sf or better	**B2** if correct answer unsupported
		i	3.5 cao	A1	**Examiner's Comments** Almost all candidates achieved full marks on this question. Some converted to degrees and rounded prematurely, thus losing the accuracy mark for the final answer, and a few used the formula for the area of a sector.	
		ii	$\cos 0.6 = \frac{d}{\text{their}3.5}$	M1	or $\cos 34.377.. = \frac{d}{\text{their}3.5}$ with θ to 3 sf or better	or correct use of Sine Rule with 0.9708(55.623°) or area = 5.709 = 0.5 × h × 3.952, or $3.5^2 - 1.976^2 = d^2$

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	2.888.. to 2.9	A1	**Examiner's Comments** This straightforward question defeated a surprisingly large number of candidates. Many of these misunderstood the question and used the Cosine Rule to calculate the length AB, or simply answered their own question and calculated the area of the sector or the segment. Many of the successful candidates used convoluted methods, such as finding AB and then using Pythagoras — premature rounding sometimes caused a mark to be lost; forgetting to halve AB cost both marks. The Sine Rule was sometimes used successfully — but this was sometimes spoiled by the use of sinπ in conjunction with 3.5. A few candidates found the area of the triangle and then used ½ base × height. Surprisingly few were able to use the expected approach: d = 3.5cos0.6.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
4		i	(A) $AC^2 = 12.8^2 + 7.5^2$ oe	M1	allow correct application of cosine rule or from finding relevant angle and using trig	
		i	$AC = 14.83543056..$	A1	rot to 3 or more sf , or 15	**B2** for 14.8 or better unsupported
		i	$\tan C = {}^{12.8}/_{7.5}$ or $C = 90 - \tan^{-1}({}^{7.5}/_{12.8})$ oe	M1	or $\sin C = {}^{12.8}/_{\text{their}14.8}$ or $\cos C = {}^{7.5}/_{\text{their}14.8}$	or $\dfrac{\sin C}{12.8} = \dfrac{\sin 90}{their 14.8}$ or $\cos C = \dfrac{their14.8^2 + 7.5^2 - 12.8^2}{2 \times 7.5 \times their 14.8}$
		i	59.6 to 59.64	A1		
		i	$\dfrac{AD}{\sin(155 - their 59.6)} = \dfrac{their 14.8}{\sin 35}$ oe	M1		

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	25.69 to 25.8	A1	allow **B2** for 25.69 ≤ *AD* < 25.8 unsupported…but **B0** for 25.8 unsupported Examiner's Comments Nearly all candidates used Pythagoras to obtain *AC* correctly. A few used the Cosine Rule instead: most were successful. However, those who did calculate arctan $\left(\frac{12.8}{7.5}\right)$ were in the minority; most used the Sine or Cosine Rule and often lost the A mark having worked with rounded or truncated values. A few used the Cosine Rule wrongly obtaining an answer close to 90° and yet failed to spot that something must be wrong. A small minority of candidates assumed that *AC* bisects angle *ACB* and a similar sized group stopped at this point. However, most went on successfully to use the Sine Rule and obtain a value within the specified range.	**M0A0** for $^{14.8}/_{\cos 55} = 25.803...$
		i	(B) area of *ABC* = 48 soi	B1	may be implied by correct final answer in range or by sight of ½ × 12.8 × 7.5 oe	condone 48.0…
		i	½×their 14.8…×their 25.7…×sin(their 59.6 – 10)	M1	may be implied by 144.8 to 146	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	192.8 to 194[m^2]	A1	<u>Examiner's Comments</u> Most candidates successfully found the area of *ABC*, although some used convoluted methods and lost accuracy or made errors with the arithmetic. Many candidates adopted the anticipated approach of ½×*AC*×*AD*×sin*DAC* and went on to present a final answer within range. However, some candidates omitted to add the two areas together and some used angle *DCA* in the formula. A small minority used convoluted methods involving the vertical height of triangle *ADC* or calculated *DC* and worked with that length instead. Accuracy was often lost, but about half of these candidates were successful.	**B3** for correct answer in range if unsupported
		ii	angle $HMG = \dfrac{\pi - 1.1}{2}$ or *MHG* = 0.55 (31.5126°)	B1	or angle *EMF* or angle *MEF*	allow 1.02 to 1.021 *or* 58.487° to 58.5°
		ii	*HM* = 1.7176 to 1.7225	B1		may be implied by final answer
		ii	½ × 1.1 × *their HM*^{2} or $\dfrac{\theta}{360} \times \pi \times theirHM^2$	M1	1.63(0661924…) θ = 63(.025357…)	check arithmetic if necessary their *HM*≠ 0.9 or 1.8
		ii	area of triangle *EMF* = 0.652 to 0.662	B1	or *MGH*	may be implied by final answer or in double this (1.304 to 1.324)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	2.95 to 2.952 [m^2] cao	A1	__Examiner's Comments__ Approximately a quarter of candidates failed to score on this question. Either no response was made, or initial assumptions such as *MH* = *MG* or *HMG* = 45° were made and no progress was made. However, most were able to obtain one of the required angles correctly and many went on to use this to find *MH* or *HG* successfully. Far too many candidates then worked with truncated values or values which were approximated too severely. The method mark for finding the area of the sector was often earned, although a few candidates used the formula for arc length, found the area of the segment or selected something more exotic from the formula booklet. A small minority converted to degrees: sometimes this was successful, but it was disappointing to see calculations such as $½ \times 1.72^2 \times 63$ on occasion. In some cases, this was added to a correct value for the triangular sections, apparently without any awareness that the numbers generated couldn't possibly match up. A number of candidates found *HG* successfully and then used Pythagoras incorrectly to obtain a value for *MH* which was smaller. Again, this was usually ignored. Approximately 20% of candidates scored full	full marks may be awarded for final answer in correct range i.e. allow recovery of accuracy

Question			Answer/Indicative content	Marks	Part marks and guidance	
					marks, but a further 7% or 8% lost the last mark either by combining their answers incorrectly or by working with rounded or truncated figures and then over-specifying their final answer.	
			Total	14		

Question			Answer/Indicative content	Marks	Part marks and guidance	
5		i	clear diagram or explanation starting with equilateral triangle correctly showing 30 as half angle and sides 1 and 2 or multiples of these lengths	B1		units for sides and angle not required
		i	correct use of Pythagoras *and* adjacent and hypotenuse correctly identified to obtain given result $\cos 30° = \frac{\sqrt{3}}{2}$	B1	adjacent and hypotenuse may be identified on diagram **Examiner's Comments** Over half of candidates failed to score on this question. A surprising number drew "equilateral" triangles with unequal angles or sides, defined the cosine ratio incorrectly or not at all, or were unable to use Pythagoras correctly to obtain the third side of their right angled triangle. Generally, candidates did not set out their work rigorously; even those who understood what was required were minimalist in their approach and missed out on both marks.	condone abbreviations
		ii	$\pm\frac{\pi}{6}$ or $-\frac{5\pi}{6}$ soi	M1	may be implied by correct answer or ±0.523598775…, or may appear on quadrant diagram or graph	condone ±30° or – 150°
		ii	$\frac{11\pi}{6}$	A1		ignore extra values outside the range

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{7\pi}{6}$	A1	if **A0A0**, **SC1** for 1.8333333π and 1.16666666π to 3 or more sf or **SC1** for 330° and 210° www __Examiner's Comments__ Almost half of candidates obtained full marks on this question. Most obtained $\pm\frac{\pi}{6}$ or $\pm 30°$ to earn the first mark; some obtained the correct angles and left their answers in degrees or only found one of the angles and a few lost a mark by adding extra values, usually $\frac{\pi}{6}$ and / or $\frac{5\pi}{6}$. Over a quarter of candidates failed to score: the usual mistake was a first move of $2\theta = \sin^{-1}(\pm 1)$.	if full marks or **SC1** awarded, subtract 1 for extra values *in* the range
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
6			71.5 (6505118..) soi	M1	or 1.24 (9045772..) (rad) or 79.5 (1672353..) (grad)	
			35.7 to 36	A1	if A0, SC1 for all four answers in radians or grad r.o.t to 3 or more sf 0.62452286, 2.195319213, 3.76611554, 5.336911867 (rad), but 0 if extra values in range	39.75836177.., 139.75…, 239.75…339.75…(grad)
			125.78…, 215.78…, 305.78… to 3 or more sf	A1	if M1A0A0, SC1 for 251.565…, 431.565…, 611.565… **Examiner's Comments** Most candidates started correctly, a few doubled 71.6 instead of halving it, but most successfully obtained 35.8°. 215.835.8° was frequently found, but the other two values were often missed. Some candidates rounded off their calculator value, and then over-specified their final values (215.79 etc was common), thus losing the second A mark. A common error was arctan(1.5) to start, and some candidates unwittingly worked in radians and went on to add multiples of 90°.	for second A1, ignore extra values outside range, A0 if extra values in range
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
7			$\dfrac{\sqrt{\sin^2\theta}}{\dfrac{\sin\theta}{\cos\theta}}$ or $\dfrac{\cos\theta\sqrt{\sin^2\theta}}{\sin\theta}$	M1	correct substitution for numerator	allow maximum of M1M1 if $\pm\sqrt{\sin^2\theta}$oe substituted
				M1	correct substitution for denominator	
			cos θ cao	A1	A0 if follows wrong working or B3 www or if unsupported **Examiner's Comments** Most candidates recognised one of the trigonometric identities required, and then made no further progress. Of those who spotted both relationships, a good proportion made a mess of simplifying the fraction, often resulting in a final answer of $\dfrac{1}{\cos\theta}$. A surprising number tried squaring top and bottom, or concocted an equation which they attempted to solve.	mark the final answer but ignore attempts to solve for θ allow recovery from omission of θ
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
8			½ × 12.4^2 × 2.1 (= 161.448)	M1*	or $\pi \times \frac{120.32}{360} \times 12.4^2$	angle in degrees to 3 sf or better
			½ × 12.4^2 × sin2.1 (= 66.3 to 66.4) or ½ × 21.5 (121..) × 6.16 (9…)	M1*	angle in degrees to 3 sf or better	may be implied by 2.81 (7168325…) (degrees) or 2.53 (5559362) (grad)
			their 161.448 – their 66.36	M1dep*		
			95 to 95.1	A1	**Examiner's Comments** This was very well done. By and large the correct formulae were used and the entire solution was worked in radians, nearly always resulting in full marks. Some candidates worked in degrees and then worked with rounded numbers, often following on to over specify their answer and lose the final mark. A significant minority did not use ½r^2½sin½03B8, but used a variety of methods in order to arrive at ½ ×base×height for the area of the triangle. Often this went astray, resulting in a loss of three marks.	if unsupported, B4 for 95.08 (446) r.o.t. to 4 sf or better
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
9			$5.9^2 + 8.5^2 - 2 \times 5.9 \times 8.5 \times \cos 72$	M1		
			107 – 31 or better	M1	76.(….) or 204.(…..) (radians)	or 64.(…..) (grad) NB 6.76 cos 72 or 2.08 (8954882..) scores M1M0
			8.7(2…)	A1	**Examiner's Comments** The cosine rule was very well understood and most candidates scored full marks. A small number left the calculator in radian mode and lost the final mark; a very small number tried to use Pythagoras or lost their way after earning the first method mark.	if M0M0, B3 for 8.72 or better if unsupported or 8.7 (2…) if obtained from other valid method
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
10		i	$[\cos A =]\dfrac{20^2+13^2-8^2}{2\times13\times20}$	M1*	or $8^2 = 20^2 + 13^2 - 2 \times 13 \times 20 \times \cos A$	
		i	$[\cos A =]\dfrac{505}{520}$ oe soi	A1	or 0.971 to 0.9712	
		i	A = 13.79 to 13.8° or 14°	A1	or 0.24077 to 0.241 or 0.24 (radians); allow **B3** if given to 3sf or more unsupported	or 15.32 (grad)
		i	[Area =] ½ × 20 × 13 × sin *their A*	M1dep*	or **M1** for eg ½ × 20 × 8 × sin 22.8, as long as angle calculated correctly from their A (other angles are 22.79824…° and 143.40645..° or 36.59355..°)	or $\sqrt{\frac{41}{2}(\frac{41}{2}-8)(\frac{41}{2}-13)(\frac{41}{2}-20)}$ NB 13sin A = 3.099899192 if ½ × b × h used

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	30.99 to 31.01 isw	A1	allow **B2** for unsupported answer within range **Examiner's Comments** Most candidates used the Cosine rule correctly to calculate angle A and most went on to calculate the area correctly using ½absinC. A minority complicated matters by splitting the triangle into 2 right angle triangles, calculating the height and then using Pythagoras to calculate the base of each triangle and hence the area of each separately. A few candidates unnecessarily found one of the other angles and then used ½absinC. In both cases this usually resulted in a loss of accuracy so their final answer was outside the permissible range. Similarly, some candidates rounded their value for cosA and went on to lose both accuracy marks. Approximately three quarters of candidates scored full marks.	
		i	or $\frac{5\sqrt{615}}{4}$ oe isw			
		ii	*alternatively*			
		ii	$h = 4$ soi	B1		
		ii	attempt to find all y-values	M1	$y_{upper} - y_{lower}$	**M0** if values are added to obtain 0.60, 0.80 etc
		ii	2.3, 2.32, 1.82, 1.34	A1	all y-values correct	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\frac{\text{their } 4}{2} \times (0 + 0 + 2(2.3{+}2.32{+}1.82{+}1.34))$	M1	shape of formula correct with 2, 3 or 4 of their *y* -values in inner bracket with their *h*; allow recovery from bracket errors **M0** if any non-zero *x*-values used or if *y*-values used twice	eg ½ × 4 × {2.3 + 1.34 + 2 (2.32 + 1.82)}
		ii		B1FT	all their *y*-values correctly placed, condone omission of zeros and/or omission of outer brackets	
		ii	31.12	A1	ignore subsequent rounding, but **A0** if answer spoiled by eg multiplication by 20 **Examiner's Comments** Most candidates used the trapezium rule separately for the upper and lower areas, with a smaller number recognising that the total area could be calculated with a single application. The most common error was omitting the outside pair of brackets in the formula and this was rarely recovered. Another common error was the failure to recognise that the absolute areas should be summed, with candidatessubtracting the lower area from the upper. A surprising error for several candidates was getting the value of h incorrect as this was not only clear from the table, but also explicitly stated in the question. Nevertheless, a significant majority scored full marks.	or **B1M1A1** + **B3** if area of 2 triangles and 3 trapezia calculated to give correct answer www NB 4.6 + 9.24 + 8.28 + 6.32 + 2.68

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	11	

Question			Answer/Indicative content	Marks	Part marks and guidance	
11			$1 - \cos^2 x = 3\cos x - 2$ oe	M1*		
			$\cos^2 x + 3\cos x - 3$ [= 0]	M1*dep	or $-\cos 2x - 3\cos x + 3 = 0$	condone one sign error *or* constant term of – 1 (in LH version) or + 1 (in RH version)
			$\cos x = \text{their } \dfrac{-3+\sqrt{21}}{2}$ or $\cos x$ = their 0.79 to 0.7913 soi	M1	dependent on award of previous method mark, must be correct for their quadratic	ignore other values (eg –3.79…); $x = 0.791287847$…but **M0** if no recovery
			[x =] 0.6578 to 0.66 isw cao	A1	**A0** for eg 0.66π if 0.66 not seen separately	NB $x = 0.65788395$…
			[x =] 5.625 to 5.63 isw cao	A1	if **A1A1** extra values in range incur a penalty of 1; ignore extra values outside range if **A0A0** allow **SC1** for 37.69 to 37.7° *and* 322 to 322.31° *or* for (0.209 to 0.21)π *and* (1.79 to 1.791)π **Examiner's Comments** A few candidates were unable to eliminate $\sin^2\theta$ legitimately, but all bar the weakest candidates managed at least 2 marks here. A small numberof candidates made errors when rearranging to zero – generally with the constant term. Some were using x for $\cos x$ in their quadratic formula and not recovering the 'cos'. This was unfortunate. Candidates must realise that this is not a useful practice. Even those who made other substitutions often failed to give their evaluated formula a subject	NB $x = 5.625301357$… no **SC** mark available if extra values in range

Question			Answer/Indicative content	Marks	Part marks and guidance	
					and then confused themselves. Some candidates resorted to rounded decimals very quickly and made premature approximation errors in their answers, thus losing one or more accuracy mark. Quite a few candidates were finding the second angle by adding 0.66 to 1.5π rather than subtracting it from 2π. A fair number of candidates worked in degrees – a good number of these were allowed the SC1 for a pair of correct answers. When the question stipulates angles over a range such as “between 0 and 2π”, the expectation is that their angleswill be in radians, not degrees.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
12			$27 = ½\ r^2 \times 1.5$ oe	M1	Or $27 = \frac{85.943669...}{360} \times \pi r^2$	angle in degrees rounded to 2 sf or more
			r = 6 soi	A1	may be embedded in formula for arc length	may be implied by later work eg 9 or 21
			their *r* × 1.5	M1	or their $\frac{85.943639}{360} \times 2\pi \times$ their *r*	if *r* is incorrect, we must **see** their *r* × 1.5 [+ 2*r*] for **M1** if *r* is correct, **M1** may be implied by 9 or 21
			21 [cm] cao	A1	allow full marks for recovery from working with rounded value of θ in degree form **Examiner's Comments** This was very well done: approximately two thirds of candidates obtained full marks. Some candidates converted to degrees and lost the accuracy marks and a few candidatesused incorrect formulae.	**B4** for 21 unsupported www
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	substitution of $\tan x = \frac{\sin x}{\cos x}$ or $\sqrt{1-\sin^2 x} = \sqrt{\cos^2 x}$ or $\cos x$ in given LHS	M1	if no substitution, statements must follow a logical order and the argument must be clear; if one substitution made correctly, condone error in other part of LHS	condone omission of variable throughout for **M1** only, but allow recovery from omission of variable at end
		i	both substitutions seen and completion to sin x as final answer	A1	**NB AG**; answer must be stated	**M0** if first move is to square one or both sides
		i			allow consistent use of other variable eg θ for both marks **Examiner's Comments** A significant minority of candidates chose to work backwards, but few were successful. Many candidates "started at both ends" and tried to meet in the middle – sometimes a method mark was achieved. A good number of candidates earned the first method mark with one of the correct substitutions, but either failed to complete the argument or tried to show something else.	Simply stating eg $\tan x = \frac{\sin x}{\cos x}$ is insufficient *Alternatively* **SC2** for complete argument eg $\tan x = \frac{\sin x}{\cos x}$ [tan x × cos x = sin x] $\sin^2 x + \cos^2 x = 1$ $\cos x = \sqrt{1-\sin^2 x}$ $\tan x = \frac{\sin x}{\sqrt{1-\sin^2 x}}$ $\tan x \times \sqrt{1-\sin^2 x} = \sin x$ oe
		ii	0, 180, 360	B1	all 3 required	**NB** sin y = 0 or ¼
		ii	14 or 14.47 to 14.5	B1	radians: mark as scheme but deduct one from total	ignore extra values outside range

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	166 or awrt 165.5	B1	0, π, 2π; 0.25 or 0.253 or awrt 0.2527; 2.89 or 2.889 or awrt 2.8889 **Examiner's Comments** Most candidates solved the quadratic successfully and went on to find 14.5 and 166. A surprising number omitted one or more of the three other roots, however.	if **B3**, deduct 1 mark for extra values within range
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	$\frac{1}{2}r^2\theta$ or $\frac{1}{2}a^2\sin\theta$ or $a^2\sin\frac{1}{2}\theta\cos\frac{1}{2}\theta$ seen	M1	do not allow use of variable other than θ	allow eg
		i	$\frac{1}{2}r^2\theta - \frac{1}{2}a^2\sin\theta$ isw oe	A1	**Examiner's Comments** This was generally very well done, but some candidates gave the area of the triangle as ½a^2 and a few gave the area of the sector as $r\theta$.	$\frac{\theta}{2\pi}\times\pi r^2$ or $\frac{1}{2}a^2\sin\left(\frac{180\theta}{\pi}\right)$ seen oe
		ii	$\frac{1}{2}a^2\sin 0.8 = \frac{1}{2}\times 12^2\times 0.8 - \frac{1}{2}a^2$ sin 0.8 oe	B1	or eg $\frac{1}{2}a^2\sin 0.8 = \frac{1}{4}\times 12^2\times 0.8$ [= 28.8] equivalent in degrees NB θ = 45.8366236…°	**NB** $a^2 = \frac{57.6}{0.717356} = 80.29485$ NB θ = 45.83662361…°
			[a =] 8.96 cao; mark the final answer	B1	if unsupported, allow **B2** for 8.96 or allow **B1** for 9.0 or 8.96074…to 4 sf or more **Examiner's Comments** A significant minority were unable to make progress with this part due to incorrect work in part 9(i). Many others set the area of the sector equal to the area of the triangle and failed to score. A few needlessly converted to degrees, and often went wrong either by losing the accuracy mark or making a method error in the formula for the sector. A surprising number of candidates ignored their correct work in part (i) and began again with incorrect expressions.	**NB** $\frac{1}{2}\sin 0.8 = 0.35867..$
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
15			$5.6^2 + 7.2^2 - 2 \times 5.6 \times 7.2 \times \cos 68$ seen	M1	may be implied by 53 or BC in range	
			53 or 53.0	M1	may be implied by BC in range	**NB** 52.9917243; (allow 47.7 to 47.71 from calculator in radian mode; may be implied by 6.90 to 6.91)
			[BC =] 7.3 or 7.27 to 7.28	A1	**NB** 7.27954…	
			$\sin C = \frac{7.2 \times \sin 68}{\text{their } BC}$	M1	or $[\cos C] = \frac{\text{their } BC^2 + 5.6^2 - 7.2^2}{2 \times 5.6 \times \text{their } BC}$	
			66 or awrt 66.5	A1	allow 1.2 or awrt 1.16 (radians); **A0** for eg 1.2 degrees	**NB** sin *C* = 0.917053… cos *C* = 0.398766…
			Alternatively eg if the perpendicular from B to AC, BX, is used			eg if perpendicular from C to AB, CY, is used, mark as follows
			7.2 × cos 68 seen	M1*	if unsupported, **B2** for 2.70 or better	5.6 × cos 68 seen
			2.7 or 2.697 to 2.70	A1		2.1 or 2.097 to 2.10
			XC = 5.6 – their AX	M1dep*	**NB** 2.902832527	BY = 7.2 – their AY
			$\tan C = \left[\frac{\text{BX}}{\text{XC}}\right] = \frac{7.2 \times \sin 68}{\text{their XC}}$	M1		$\tan B = \left[\frac{\text{CY}}{\text{BY}}\right] = \frac{5.6 \times \sin 68}{\text{their BY}}$
			66 or awrt 66.5	A1	allow 1.2 or awrt 1.19 (radians); **A0** for eg 1.2 degrees **Examiner's Comments** This was very well done; the majority of candidates obtained full marks and almost all achieved at least 4 marks. A few worked with rounded numbers and then over-specified their final answer, thus losing the final accuracy mark, and a few left their calculator in radian mode and usually lost both accuracy marks.	C [= 90 – B] = 66 or awrt 66.5
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$a = 2$, $b = ½$	B1B1	**Examiner's Comments** For some, this was a write-down for 2 marks. Some used transformation arguments, other substituted in particular coordinates. The most common errors were a = 3 and b = 2 or ¼ .	
		ii	$y = 2 + \cos ½ x \leftrightarrow x \leftrightarrow y$ $x = 2 + \cos ½ y$		(may be seen later)	
		ii	$\Rightarrow x - 2 = \cos ½ y$	M1	subtracting [their] a from both sides (first)	need not substitute for a, b
		ii	$\Rightarrow \arccos (x - 2) = ½ y$	M1	arccos (x –[their] a) = [their] $b \times y$	or with $x \leftrightarrow y$, need not subst for a, b
		ii	$\leftrightarrow y = f^{-1} (x) = 2 \arccos (x - 2)$	A1	cao or $2 \cos^{-1} (x - 2)$	may be implied by flow diagram
		ii	Domain $1 \leq x \leq 3$	M1	domain 1 to 3, range 0 to 2 π	if not stated, assume first is domain
		ii	Range $0 \leq y \leq 2\pi$	A1	correctly specified: must be ≤, x for domain, y or f^{-1} or $f^{-1}(x)$ for range **Examiner's Comments** We allowed plenty of follow-through marks for the first two method marks here, which were almost always gained. Fully correct inverse functions were common. As for domain and range, most seem to know that the these are the reverse of the domain and range for f. However, the 'A' mark here demanded accurate use of notation, with 'x' used for domain and 'y' for range, and this mark was often lost.	allow [1, 3], [0, 2 π] not 360° (not f)

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	7		
17		i	arcsin $x = \pi/6 \Rightarrow x = \sin \pi/6$	M1		
		i	= ½	A1	allow unsupported answers **Examiner's Comments** This was an easy two marks for virtually all candidates, though occasionally they lefttheir answer as sin π/6 without evaluating this as ½.	
		ii	sin π/4 = cos π/4 = 1/√ 2			
		ii	⇒ arcsin (1/√ 2) = arccos (1/√ 2)			
		ii	⇒ x = 1/√ 2	B2	o.e. e.g. √ 2/2, must be exact; SCB1 0.707… **Examiner's Comments** This part was the polar opposite of part (i), with very few candidates getting anywhere. Two common errors were to infer that sin x = cos x and therefore x = π/4, and dividing arc sin x by arc cosx to get arc tan x. Successful candidates usually introduced another variable y equal to arc sin x, so that sin y = cos y, tan y = 1, y = π/4, and x = sin π/4 = 1/√2.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
18		i	AC = cosecθ	M1	or 1/sinθ	
		i	⇒ AD = cosecθ sec φ	A1	oe but not if a fraction within a fraction **Examiner's Comments** This question was answered well by the most able but many others could not cope with the fractions in part (i). AC was generally correct but often AD=cosϕ/sinθ or sinθ/cosϕ was the given answer, whilst others left their answer as a fraction within a fraction.	
		ii	DE = AD sin (θ + φ)		AD sin(θ + φ) with substitution for their AD correct compound angle formula used **Do not** award both M marks unless they are part of the same method. (They may appear in either order.) simplifying using tan = sin/cos. A0 if no intermediate step as **AG** ----------------------------- from triangle formed by using X on DE where CX is parallel to BE to get DX = CD cosθ and CB = 1 (oe trigonometry) substituting for both CD = AD sinφ and their AD oe to reach an expression for DE **in terms of θ and φ only** (M marks must be part of same method) **AG** simplifying to required form	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	= cosecθ sec φ sin (θ + φ)	M1	AD sin(θ + φ) with substitution for their AD	
		ii	= cosecθ sec φ(sinθ cos φ + cosθ sin φ)	M1	correct compound angle formula used	
		ii	$= \frac{\sin\theta\cos\varphi + \cos\theta\sin\varphi}{\sin\theta\cos\varphi}$		**Do not** award both M marks unless they are part of the same method. (They may appear in either order.)	
		ii	$= 1 + \frac{\cos\theta}{\sin\theta}\frac{\sin\varphi}{\cos\varphi}$			
		ii	= 1 + tan φ / tan θ* ----------------------------- **OR** equivalent,	A1	simplifying using tan = sin/cos. A0 if no intermediate step as **AG**	
		ii			-----------------------------	
		ii	eg from DE = CB + CD cosθ = 1 + CD cosθ = 1 + AD sin φ cosθ	M1	from triangle formed by using X on DE where CX is parallel to BE to get DX = CD cosθ and CB = 1 (oe trigonometry)	
		ii	= 1 + cosec θ sec φ sin φ cosθ	M1	substituting for both CD = AD sinφ and their AD oe to reach an expression for DE **in terms of θ and φ only** (M marks must be part of same method)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	= 1 + tanφ / tanθ*	A1	**AG** simplifying to required form **Examiner's Comments** Good candidates were able to answer this with ease. Quite a few candidates made no response. Much depended upon their answers in part (i) which were followed through for the method marks. Those who then did not obtain the given answer should have realised that they ought to have reconsidered their answer to part (i).	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
19			$2\sec^2\theta = 5\tan\theta$	6		
			$\Rightarrow 2(1 + \tan^2\theta) = 5\tan\theta$	M1	$\sec^2\theta = 1 + \tan^2\theta$ **used**	
			$\Rightarrow 2\tan^2\theta - 5\tan\theta + 2 = 0$	A1	correct quadratic **oe**	
			$\Rightarrow (2\tan\theta - 1)(\tan\theta - 2) = 0$	M1	solving their quadratic for $\tan\theta$ (follow rules for solving as in Question 1 [*,*]	
			$\Rightarrow \tan\theta = ½$ or 2	A1	www	
			$\Rightarrow \theta = 0.464$,	A1	first correct solution (or better)	
			1.107	A1	second correct solution (or better) and no others in the range Ignore solutions outside the range. **SC** A1 for both 0.46 and 1.11 **SC** A1 for both 26.6° and 63.4° (or better) Do not award **SC**s if there are extra solutions in range.	
			---------------------------- OR		----------------------------	
			$2/\cos^2\theta = 5\sin\theta/\cos\theta$	M1	**using** both sec = 1/cos and tan = sin/cos	
			$\Rightarrow 2\cos\theta = 5\sin\theta\cos^2\theta$, $\cos\theta \neq 0$	A1	correct one line equation $2 - 5\sin\theta\cos\theta = 0$ or $2\cos\theta = 5\sin\theta\cos^2\theta$ oe (or common denominator). Do not need $\cos\theta \neq 0$ at this stage.	
			$\Rightarrow \cos\theta(2 - 5\sin\theta\cos\theta)$ 0 $\Rightarrow \cos\theta = 0$, *or* $\sin 2\theta = 0.8$	M1	**using** $\sin 2\theta = 2\sin\theta\cos\theta$ oe eg $2 = 5\sin\theta\sqrt{(1 - \sin^2\theta)}$ and squaring	
			$\Rightarrow \sin 2\theta = 0.8$	A1	$\sin 2\theta = 0.8$ or, say, $25\sin^4\theta - 25\sin^2\theta + 4 = 0$	
			$\Rightarrow 2\theta = 0.9273$ *or* 2.2143 $\Rightarrow \theta = 0.464$	A1	first correct solution (or better)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			1.107	A1	second correct solution (or better) and no others in range Ignore solutions outside the range SCs as above **Examiner's Comments** Candidates seemed equally to choose the two approaches in the mark scheme to solve the trigonometric equation. Both were equally successful and few offered extra unnecessary solutions. The main error was to give insufficient accuracy in the final solutions. Where solving tan θ=2 in degrees leads to θ=63.4° to 3sf, giving θ= 1.11 radians = 63.598° (63.6°) and θ=1.1radians=63.0° were insufficiently accurate so we needed θ=1.107radians to achieve the same accuracy as 63.4°.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
20		i	$v\mathrm{d}v/\mathrm{d}x + 4x = 0$ $\int v\mathrm{d}v = -\int 4x\,\mathrm{d}x$	M1	separating variables and intending to integrate	
		i	$\frac{1}{2}v^2 = -2x^2 + c$	A1	oe condone absence of c. [Not immediate $v^2 = -4x^2$ (+c)]	
		i	When $x = 1$, $v = 4$, so $c = 10$	B1	finding c, must be convinced as AG, need to see at least the statement given here oe (condone change of c)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	so $v^2 = 20 - 4x^2$ *	A1	**AG** following finding *c* convincingly Alternatively, **SC** $v^2 = 20 - 4x^2$, by differentiation, $2v\,dv/dx = -8x$ $v\,dv/dx + 4x = 0$ scores B2 if, in addition, they check the initial conditions a further B1 is scored (ie 16 = 20 – 4). Total possible 3/4. <u>Examiner's Comments</u> Most candidates scored all four marks when solving the differential equation. It was pleasing to see so few candidates failing to include the constant of integration. Some candidates, however, tried to work backwards from the answer, or wrote $v^2 = -4x^2 + c$ without showing from where it came. The answer was given in this case so stages of working were needed. Whilst, on this occasion, examiners condoned the change of constant candidates should be encouraged to change their constant when appropriate in the future and not use *c* twice to mean different things within the same question.	
		ii	$x = \cos 2t + 2\sin 2t$ when $t = 0$, $x = \cos 0 + 2 \sin 0 = 1$*	B1	**AG** need some justification	

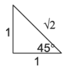
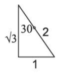
Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$v = \mathrm{d}x/\mathrm{d}t = -2\sin 2t + 4\cos 2t$	M1	differentiating, accept ±2, ±4 as coefficients but not ±1, ±2 and not ±1/2, ±1 from integrating	
		ii		A1	cao	
		ii	$v = 4\cos 0 - 2\sin 0 = 4$*	A1	www AG __Examiner's Comments__ Most candidates obtained the mark for verifying that x =1. Many others also scored the following three marks but some had the incorrect coefficients when differentiating and only had the correct coefficient in the second term when working backwards from the answer, 4.	
		iii	$\cos 2t + 2\sin 2t = R\cos(2t - \alpha) = R(\cos 2t\cos\alpha + \sin 2t\sin\alpha)$ $R = \sqrt{5}$	B1	or 2.24 or better (not ± unless negative rejected)	
		iii	$R\cos\alpha = 1$, $R\sin\alpha = 2$	M1	correct pairs soi	
		iii	$\tan\alpha = 2$,	M1	correct method	
		iii	$\alpha = 1.107$	A1	cao radians only, 1.11 or better (or multiples of π that round to 1.11)	
		iii	$x = \sqrt{5}\cos(2t - 1.107)$ $v = -2\sqrt{5}\sin(2t - 1.107)$	A1	differentiating or otherwise, ft their numerical R, α (not degrees) required form **SC** B1 for $v = \sqrt{20}\cos(2t + 0.464)$ oe	
		iii	EITHER $v^2 = 20\sin^2(2t - \alpha)$ $20 - 4x^2 = 20 - 20\cos^2(2t - \alpha)$	M1	squaring their v (if of required form with same α as x), and x, and attempting to show $v^2 = 20 - 4x^2$ ft their R, α (incl. degrees) [α may not be specified].	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$= 20(1 - \cos^2(2t - \alpha)) = 20\sin^2(2t - \alpha)$ so $v^2 = 20 - 4x^2$	A1	cao www (condone the use of over-rounded α (radians) or degrees)	
		iii	OR multiplying out $v^2 = (-2\sin 2t + 4\cos 2t)^2$ $= 4\sin^2 2t - 16\sin 2t\cos 2t + 16\cos^2 2t$ and $4x^2 = 4(\cos^2 2t + 4\sin 2t\cos 2t + 4\sin^2 2t)$ $= 4\cos^2 2t + 16\sin 2t\cos 2t + 16\sin^2 2t$ (need middle term) and attempting to show that $v^2 + 4x^2 = 4(\sin^2 2t + \cos^2 2t) + 16(\cos^2 2t + \sin^2 2t)$ $= 4 + 16 = 20$ (or $20 - 4x^2 = v^2$) oe	M1	differentiating to find v (condone coefficient errors), squaring v and x and multiplying out (need attempt at middle terms) and attempting to show $v^2 = 20 - 4x^2$	
		iii	so $v^2 = 20 - 4x^2$	A1	cao www	
		iii			Rcosα = 1, Rsinα = 2, R = √5,α = 1.107 1) Missing R Ie cosα = 1, sinα = 2. We reluctantly condone this provided that it is followed by working that suggests R was implied such as tanα = 2, α = 1.107 M1M1A1. Other methods are possible. We do not award M1 for cosα = 1, sinα = 2 if it is followed by α = inv cos 1 as R is not implied. B1 is still available. 2) Incorrect pairs eg Rsinα = 1, Rcosα = 2 scores M0 but would obtain the second M1ft if it was followed by tanα = 1/2. M0M1A0. B1 is possible. 3) Incorrect method Rsinα = 2, Rcosα = 1 followed by tanα = 1/2 scores M1M0A0. B1 is possible.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					4) Incorrect pairs and incorrect method Rsinα = 1, Rcosα = 2, tanα = 2 is M0M0A0. B1 is possible. This is easily over-looked and is a double error leading to an apparently correct answer. 5) Incorrect signs (all could score B1) (a) Rcosα = 1, Rsinα = –2, tanα = –2, M1, M1ft, A0 (b) Rcosα = 1, Rsinα = –2, tanα = 2, M1 M0ft, A0 sign error (c) Rcosα = 1, Rsinα = –2, R = √5, sinα = –2/√5, M1 M1 A0 sign error (d) Rcosα = 1, Rsinα = –2, sinα = 2/√5, M1M0 sign error A0 (e) Rcosα = 1, Rsinα = –2, cosα = 1/√5, α = 1.107 M1, M1, A0 sign error (even though not used) 6) Incorrect R a) Rcosα = 1, Rsinα = 2, R = 5 (say), cosα = 1/5, scores B0 M1M1ftA0 b) Rcosα = 1, Rsinα = 2, R = 5 (say), tanα = 2, α = 1.107 scores M1M1B0A1 (allow) 7) Missing Working a) tanα = 2, α = 1.107, R = √5, scores M1M1A1B1 soi b) tanα = 1/2, R = √5 scores M1M0B1A0 (either correct pairs or correct method but not both) c) Rcosα = 1, R = √5, α = 1.107 M1M1B1A1 soi Other options are possible. Examiners should consult their Team Leaders if in doubt.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					<u>Examiner's Comments</u> This part was rarely answered completely successfully. Most candidates understood that the '*R*' method was needed and scored the first three marks. This was the modal mark. Calculus was needed in this question. The candidates were asked to find the constant *α*, and *t* is a time to be combined in (2*t*-*α*) so answers given in radians were required. The use of degrees here was a very common mistake. Many candidates then differentiated their angles in degrees and obtained no marks for *v*. The last two marks were obtained only rarely but they were a good differentiator for the able candidates. Use of unspecified α, or in degrees or radians was allowed in this last part. Some candidates had difficulty as they had found both *x* and *v* separately using the '*R*' method and so had different values for their angles. Others realised the problem and were able to use trigonometric identities to change their *v* (or *x*) to have the same angle.	
		iv	$x = \sqrt{5}\cos(2t - \alpha)$ or otherwise x max = $\sqrt{5}$	B1	ft their *R*	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	when $\cos(2t - \alpha) = 1$, $2t - 1.107 = 0$,	M1	oe (say by differentiation) ft their α in radians or degrees for method only	
		iv	$2t = 1.107$ $t = 0.55$	A1	cao (or answers that round to 0.554) **Examiner's Comments** The majority of candidates scored the first two marks.. It was disappointing that candidates did not realise their mistake in part (ii) when they obtained an answer of time, t= 31.7 degrees.	
			Total	18		

Question	Answer/Indicative content	Marks	Part marks and guidance
21	 tan 45° = 1/1 = 1*  tan 30° = 1/√3*		For both B marks **AG** so need to be convinced and need triangles but further explanation need not be on their diagram. Any given lengths must be consistent.
		B1	Need √2 or indication that triangle is isosceles oe
		B1	Need all three sides oe
	tan 75° = tan (45° + 30°)	M1	use of **correct** compound angle formula with 45°, 30° soi
	$= \frac{\tan 45 + \tan 30}{1 - \tan 45 \tan 30} = \frac{1 + 1/\sqrt{3}}{1 - 1/\sqrt{3}}$	A1	substitution in terms of √3 in any **correct** form
	$= \frac{1+\sqrt{3}}{-1+\sqrt{3}}$ $= \frac{(1+\sqrt{3})^2}{3-1}$	M1	eliminating fractions within a fraction (or rationalising, whichever comes first) provided compound angle formula is used as tan(A + B) = tan(A ± B)/(1 ± tan$A$tan$B$).
	(oe eg $\frac{3+\sqrt{3}}{3-\sqrt{3}} = \frac{(3+\sqrt{3})^2}{9-3}$)	M1	rationalising denominator (or eliminating fractions whichever comes second)
	$= \frac{(3+2\sqrt{3}+1)}{3-1} = 2+\sqrt{3}$ *	A1	**correct** only, **AG** so need to see working
			Examiner's Comments There were some good explanations with appropriate triangles in the first part. However, too many candidates felt it was enough to only give the information given in the question and this was not sufficient. More was needed

Question			Answer/Indicative content	Marks	Part marks and guidance	
					than, for example, a right-angled triangle with lengths of 1, 1 and 45° to show that tan 45°=1. It was necessary to clearly show the triangle was isosceles by giving the other angle or showing that the hypotenuse was √2, or equivalent. Some made errors when calculating the other lengths in both triangles. Some good candidates failed to score here seemingly being unfamiliar with where these identities came from. The second part started well for most candidates, who usually used the correct compound angle formula, (although there were a few who thought that tan75°=tan45°+tan30°) and made the first substitution. Thereafter, this question gave the opportunity for candidates to show that they could eliminate fractions within fractions and rationalise the denominator. This was a good discriminator for the higher scoring candidates. A few candidates abandoned their attempt at half way and equated $\frac{1+\frac{1}{\sqrt{3}}}{1-\frac{1}{\sqrt{3}}}$ at that stage to the given answer $2+\sqrt{3}$.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
22			cosec x + 5 cot x = 3 sin x			
			$\Rightarrow \frac{1}{\sin x}+\frac{5\cos x}{\sin x}=3\sin x$	M1	**using** cosec x = 1/sin x and cot x = cos x / sin x	
			$\Rightarrow$ 1 + 5 cos x = 3 $\sin^2 x$ = 3(1 – $\cos^2 x$)	M1	$\cos^2 x + \sin^2 x = 1$ **used** (both M marks must be part of same solution in order to score both marks)	
			$\Rightarrow$ 3 $\cos^2 x$ + 5 cos x – 2 = 0 *	A1	**AG** (**Accept working backwards, with same stages needed**)	
			$\Rightarrow$ (3 cos x – 1)(cos x + 2) = 0	M1	use of correct quadratic equation formula (can be an error when substituting into correct formula) or factorising (giving correct coeffs 3 and –2 when multiplied out) or comp square oe	
			$\Rightarrow$ cos x = 1/3,	A1	cos x = 1/3 www	
			x = 70.5°,	A1	for 70.5° or first correct solution, condone over-specification (ie 70.5° or better eg 70.53°, 70.5288° etc),	
			289.5°	A1	for 289.5° or second correct solution (condone over-specification) and no others in the range Ignore solutions outside the range **SC**A1A0 for incorrect answers that round to 70.5 and 360-their ans, eg 70.52 and 289.48 **SC** Award A1A0 for 1.2, 5.1 radians (or better) Do not award SC marks if there are extra solutions in the range <u>Examiner's Comments</u>	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					Many candidates scored full marks when showing that the trigonometric equation could be rearranged as a quadratic and then solving it. Where there were errors, these were usually in the first part when trying to establish the given result. Errors included failing to use the correct trigonometric identities, failing to use $\sin^2\theta + \cos^2\theta = 1$ or squaring the original expression term by term. Few candidates would say $x+3=7$ so $x^2+9=49$ and yet they happily square cosec $x+5\cot x=3\sin x$ term by term. Those who were unable to complete the first part sensibly then proceeded to solve the quadratic equation. Few errors were seen here. Occasionally the final solution was incorrect and few candidates offered additional incorrect solutions.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
23		i	EITHER Use of cos=1/sec (or sin= 1/cosec)	3	Must be used	
		i	intermediate steFrom RHS	B1		
		i	$\frac{1-\tan\alpha\tan\beta}{\sec\alpha\sec\beta}$ $=\frac{1-\sin\alpha/\cos\alpha.\sin\beta/\cos\beta}{1/\cos\alpha.1/\cos\beta}$ $=\cos\alpha\cos\beta(1-\frac{\sin\alpha\sin\beta}{\cos\alpha\cos\beta})$			
		i		M1	Substituting and simplifying as far as having no fractions within a fraction [need more than $\frac{1-tt}{\sec\sec}=cc-ss$ ie an intermediate step that can lead to cc-ss]	
		i	= cos α cos β – sin α sin β			
		i	= cos(α + β)	A1	Convincing simplification and correct use of cos(α + β) **Answer given**	
		i	OR From LHS, cos = 1/sec or sin = 1/cosec used $\cos(\alpha+\beta)$ $=\cos\alpha\cos\beta-\sin\alpha\sin\beta$ $=\frac{1}{\sec\alpha\sec\beta}-\sin\alpha\sin\beta$	B1		
		i	$=\frac{1-\sec\alpha\sin\alpha\sec\beta\sin\beta}{\sec\alpha\sec\beta}$	M1	Correct angle formula and substitution and simplification to one term OR eg cosαcosβ – sinαsinβ = cos α cos β (1 – tan α tan β)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$= \dfrac{1 - \tan\alpha\tan\beta}{\sec\alpha\sec\beta}$	A1	Simplifying to final answer www **Answer given** Or any equivalent work but must have more than cc-ss = answer. **Examiner's Comments** There were some very good solutions here when showing the two trigonometric expressions were equal. However, the majority were not successful. The most common overall error was not treating both sides of an equation equally. Too often only one side was changed. A common starting point was $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta=\cos\alpha\cos\beta-\sin\alpha\sin\beta=1-\tan\alpha\tan\beta$. This was then followed by a confused attempt at dividing by $\sec\alpha\sec\beta$. Candidates need to multiply 'top and bottom' by the same thing. Questions that involve 'Showing' need more rigour.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\beta = \alpha$	2	$\beta = \alpha$ used, Need to see $\sec^2\alpha$ Use of $\sec^2 \alpha = 1 + \tan^2\alpha$ to give required result **Answer Given** Use of $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$ soi Simplifying and using $\sec^2 \alpha = 1 + \tan^2 \alpha$ to final answer **Answer Given** Accept working in reverse to show RHS = LHS, or showing equivalent	
		ii	$\cos 2\alpha = \dfrac{1 - \tan^2 \alpha}{\sec^2 \alpha}$	M1	$\beta = \alpha$ used, Need to see $\sec^2\alpha$	
		ii	$= \dfrac{1 - \tan^2 \alpha}{1 + \tan^2 \alpha}$	A1	Use of $\sec^2 \alpha = 1 + \tan^2\alpha$ to give required result **Answer Given**	
		ii	OR, without Hence,			
		ii		M1	Use of $\cos 2\alpha = \cos^2 \alpha - \sin^2 \alpha$ soi	
		ii	$h = (1 - \frac{1}{2}At)^2$	A1	Simplifying and using $\sec^2 \alpha = 1 + \tan^2 \alpha$ to final answer **Answer Given** Accept working in reverse to show RHS = LHS, or showing equivalent **Examiner's Comments** This part was more successful provided that candidates wrote down the identity for $\sec^2\alpha$. There were, however, some long and confused attempts.	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$\cos 2\theta = \frac{1}{2}$	M1	Soi or from $\tan^2\theta = 1/3$ oe from $\sin^2\theta$ or $\cos^2\theta$	
		iii	i. $2\theta = 60°, 300°, \theta = 30°,$	A1	First correct solution	
		iii	150°	A1	Second correct solution and no others in the range SC B1 for $\pi/6$and $5\pi/6$ and no others in the range **Examiner's Comments** Most candidates scored the first two marks here. Many failed to give the second solution of 150°.	
			Total	8		

Question			Answer/Indicative content	Marks	Part marks and guidance	
24		i	$\cos x + \lambda \sin x = R\cos(x - \alpha)$	Enter text here.	Enter text here.	
		i	$= R\cos x \cos\alpha + R\sin x \sin\alpha$		Enter text here.	
		i	$\Rightarrow R\cos\alpha = 1$, $R\sin\alpha = \lambda$	M1	Correct pairs. Condone sign error (so accept $R\sin\alpha = -\lambda$)	
		i	$\Rightarrow R^2 = 1 + \lambda^2$, $R = \sqrt{(1 + \lambda^2)}$	B1	Positive square root only – isw. Accept $R = 1/\cos(\arctan\lambda)$or $R = \lambda/\sin(\arctan\lambda)$	
		i	$\tan\alpha = \lambda$ (oe)	M1	Follow through their pairs. $\tan\alpha = \lambda$ with no working implies both M marks. However, $\cos\alpha = 1$, $\sin\alpha = \lambda \Rightarrow \tan\alpha = \lambda$ scores M0M1. First two M marks may be implied by combining one of the pairs with R eg, $\cos\alpha = \frac{1}{\sqrt{(1+\lambda^2)}}$ or $\sin\alpha = \frac{\lambda}{\sqrt{(1+\lambda^2)}}$	
		i	$\Rightarrow \alpha = \arctan\lambda$ (oe)	A1	$\alpha = \arccos\left(\frac{1}{\sqrt{(1+\lambda^2)}}\right)$, $\alpha = \arcsin\left(\frac{\lambda}{\sqrt{(1+\lambda^2)}}\right)$ Accept embedded answers, eg, $\sqrt{(1 + \lambda^2)}\cos(x - \lambda)$ for full marks	
		ii	max is R so $R = 2$	B1	Enter text here.	
		ii	$1 + \lambda^2 = 4 \Rightarrow \lambda = \sqrt{3}$	M1 A1	M1 for using $\sqrt{(1 + \lambda^2)} = R_{max}$, A0 for $\pm\sqrt{3}$ as final answer	
		ii	$\alpha = \arctan\sqrt{3} = \pi/3$	B1	www (eg $\lambda = 1$ and $\cos\alpha = (1 + \lambda)^{-1} \Rightarrow \alpha = \pi/3$ is B0) Exact answers only for final A and B marks **Examiner's Comments** This question differentiated well due to the coefficient of sin x taking the form of a	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					positive constant rather than a number. Many candidates, however, were unfazed by this and worked out the correct values for R and α . Some candidates lost the first method mark by not including R inthe expanded trigonometric statements $R\cos\alpha = 1, R\sin\alpha = \lambda$. Writing α in terms of the more complex arcsin and arccos expressions was surprisingly common. It was a littleworrying that a sizeable minority of candidates went from the correct $R = \sqrt{1+\lambda^2}$ to the incorrect $R = 1 + \lambda$, thinking the squared terms and the square root cancelled each other out. In part (ii) those candidates that realised that $R = 2$ usually went on to get the correct values for λ and α. However it was common for λ to be incorrect due to an incorrect expression for R from part (i). A fair proportion of candidates gave α in degrees and those who gave α as either arccos $\left(\frac{1}{\sqrt{1+\lambda^2}}\right)$ or arcsin $\left(\frac{\lambda}{\sqrt{1+\lambda^2}}\right)$ were generally less successful in this part than those who gave α as arctanλ.	

Question			Answer/Indicative content	Marks	Part marks and guidance
			Total	8	

Question			Answer/Indicative content	Marks	Part marks and guidance	
25			$\cos 2\theta = 1 - 2\sin^2\theta$	M1*	$\cos 2\theta = \pm 1 \pm 2\sin^2\theta$ (maybe implied in substitution)	
			$(6\cos 2\theta + \sin\theta =)\ 6 - 12\sin^2\theta + \sin\theta$	A1		
			$6\cos 2\theta + \sin\theta = 0$ $\Rightarrow 12\sin^2\theta - \sin\theta - 6 = 0$ $\Rightarrow (4\sin\theta - 3)(3\sin\theta + 2) = 0$	M1dep*	Use of correct quadratic equation formula or factorising or comp. the square on their three term quadratic in $\sin\theta$ (see guidance in question 1 for awarding this method mark) provided $b^2 - 4ac \geq 0$	
			$\Rightarrow \sin\theta = 3/4$ or $-2/3$	A1	www	
				B1	First correct solution to 1 dp or better (eg 48.59° etc)	
				B1	Three correct solutions	
			$\Rightarrow \sin\theta = 3/4,\ \theta = 48.6°, 131.4°$ $\sin\theta = -2/3,\ \theta = 221.8°, 318.2°$	B1	All four correct solutions and no others in the range Ignore solutions outside the range **SC** Award max B1B1B0 for answers in radians (0.85, 2.29, 3.87, 5.55 or better – so one correct B1, three correct B1). Award max B1 if there are extra solutions in the range with radians **SC** If M1M1 awarded and both values of $\lvert\sin\theta\rvert \leq 1$ but B0B0B0 then award B1 only for evidence of using $\sin\theta \equiv \sin(180 - \theta)$ **Examiner's Comments** The majority of candidates correctly replaced $\cos 2\theta$ with $1 - 2\sin^2\theta$ although a minority of candidates made the costly mistake of replacing $\cos 2\theta$ with 1 –	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					$\sin^2\theta$. While some candidates struggled to factorise $12\sin^2\theta - \sin\theta - 6 = 0$ many used the quadratic formula to solve this equation, and as withquestion 1, there were some candidates who did not state or apply the quadratic formula correctly. While the majority of candidates found the correct values forsinθ some incorrectly obtained $-\frac{3}{4}$ and/or $\frac{2}{3}$. Of those candidates that obtained the correct values for sinθ the majority went on to score full marks. However, it was fairly common to see 'arcsin $\left(-\frac{2}{3}\right)$ = –41.81… *therefore no solutions in the range*' with no appreciation that solutions in the correct range could be found from this value. Having found the principal values it was common for candidates to get the other solutions in the range, often sketching the sine curve to help them, though most did this correctly without demonstrating any method.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
26			$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t}=\frac{-2/t^2}{2}\left(=-\frac{1}{t^2}\right)$	M1*	M1 for their (dy / dt) / their (dx / dt) in terms of t with at least one term correct	
				A1	A1 cao (oe) – allow unsimplified even if subsequently cancelled incorrectly i.e. can isw	
				M1dep*	$y-\frac{2}{t}=\mathrm{f}(t)(x-2t)$ with any non-zero gradient expressed as a function of t - or any equivalent form (e.g. $y = mx + c$) but must have used the correct point $\left(2t,\frac{2}{t}\right)$ - if using $y = mx + c$ must explicitly have $c = \ldots$ before M1 can be awarded	
			$y-\frac{2}{t}=-\frac{1}{t^2}(x-2t)$	A1	oe - need not be simplified	
			When $x=0,\ y=\frac{4}{t}$	A1ft	Must be a function of t	
			When $y = 0$, $x = 4t$	A1ft	Must be a function of t	
			So area of $\text{triangle}=\frac{1}{2}\times\frac{4}{t}\times 4t=8$ (which is independent of t)	A1	No ft on this mark – an answer of 8 (www) with no additional comment is sufficient to award this mark	
			OR (for the first two marks) $y=\frac{2}{\left(\frac{x}{2}\right)}=\frac{4}{x}\Rightarrow\frac{\mathrm{d}y}{\mathrm{d}x}=-\frac{4}{x^2}$	M1*	Attempt to eliminate t and correctly differentiates their Cartesian equation	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			$\Rightarrow \left(\frac{dy}{dx}\right) = -\frac{4}{(2t)^2}$	A1	**Examiner's Comments** Candidates found this unstructured question on parametric equations quite demanding with the modal mark scored being 2 out of a possible 7. Most candidates scored these two marks by correctly obtaining $\frac{dy}{dx} = -\frac{1}{t^2}$ although the vast majority failed to make any further progress worthy of merit with many trying to argue that the area of triangle OQR is independent of t without any further working of a mathematical nature. Of those that failed to obtain the correct derivative a number incorrectly stated that $\frac{dy}{dt} = 2\ln t$ or implied that $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dx}{dt}$. All that was required from those candidates who had obtained the correct gradient function in terms of t was to write down the equation of the tangent $\left(y - \frac{2}{t} = -\frac{1}{t^2}(x - 2t)\right)$, substitute $x = 0$ and $y = 0$ to obtain $R\left(0, \frac{4}{t}\right)$ and Q($4t$,0) respectively and hence calculate the area of the triangle as	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					$8\left(=\frac{1}{2}\times 4t\times\frac{4}{t}\right)$ which is clearly independent of t. A number of candidates began by finding the Cartesian equation of the curve and correctly obtaining $\frac{dy}{dx}=-\frac{4}{x^2}$ but they then incorrectly used this gradient function in their equation of the tangent.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
27		i	AC = $x \sec \theta$	B1	Accept any equivalent form (e.g. AC $\cos \theta = x$). If AC not seen then there must be a diagram as evidence of correct sides - $x\sec \theta$ with no AC is B0	
		i	AD = $x \sec^2\theta$ and AE = $x \sec^3\theta$	B1	Accept $2x = x \sec^3\theta$ (as AE = $2x$) or any equivalent form. Otherwise there must be a corresponding diagram as evidence of correct sides. Accept $\cos^3\theta = x / \mathrm{AC} \times \mathrm{AC} / \mathrm{AD} \times \mathrm{AD} / 2x$ for the first two marks	
		i	$\Rightarrow x \sec^3\theta = 2x$ $\Rightarrow \sec^3\theta = 2$*	B1	This line (oe) must be seen before the x's cancelled **NB AG** – dependent on all previous marks	
		i	OR AD = $2x \cos \theta$	B1	Same principles as above for each corresponding mark	
		i	AC = $2x \cos^2\theta$ and AB = $2x \cos^3\theta$	B1	or $x = 2x \cos^3\theta$ (as AB = x)	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		i	$2x\cos^3\theta = x \Rightarrow \sec^3\theta = 2$*	B1	Must see $2x\cos^3\theta = x$ (oe) before given answer **Examiner's Comments** This question provided a certain amount of discrimination between candidates with some producing clear, concise arguments for why $\sec^3\theta = 2$ and why the ratio of the lengths ED to CB was $2^{\frac{2}{3}}:1$ while a significant number left both parts of this question blank or scored no marks. The majority of candidates, however, scored at least one mark in (i) for starting that AC = $x\sec\theta$ (or equivalent) or that AD = $2x\cos\theta$ but many failed to find corresponding expressions for either AD and AE or AC and AB in terms of x and one of $\sec\theta$ or $\cos\theta$. Examiners noted that many candidates did not make it clear which expression corresponded to which side of the three triangles given in the question making it almost impossible for examiners to award any marks.	
		ii	ED = $2x\sin\theta$	B1	oe e.g. $\text{ED} = \sqrt{4x^2 - \text{AD}^2}$ or ED = AD tan θ with AD correctly expressed in terms of x and θ (or using θ = 37.5 or better) - see (i) for alternatives for AD. Allow ED = $1.22x$ (or better) but B0 if ED = … missing	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	CB = $x \tan \theta$	B1	oe e.g. $CB = \sqrt{AC^2 - x^2}$ or CB = AC sin θ with AC correctly expressed in terms of x and θ (or using θ = 37.5 or better) - see (i) for alternatives for AC. Allow CB = 0.77x (or better) but B0 if CB = … missing	
		ii	$\dfrac{\text{ED}}{\text{CB}} = \dfrac{2x\sin\theta}{x\tan\theta} = 2\cos\theta$	B1	www must come from exact working (so not using θ = 37.46… oe) - accept $\frac{\text{ED}}{\text{CB}} = \frac{2}{\sec\theta}$ or $\frac{\text{ED}}{\text{CB}} = \sec^2\theta$ (oe) (as from (i): $\sec^3\theta = 2$)	
		ii	$= 2/2^{\frac{1}{3}} = 2^{\frac{2}{3}}$ *	B1	**NB AG** – dependent on all previous marks in (ii) – must be one step of intermediate working from 2cos θ to given answer **Examiner's Comments** many candidates scored at least two marks for stating that ED = 2$x$sinθ and CB = $x$tanθ although many then substituted in the angle from part (i) and tried to derive the exact value of $2^{\frac{2}{3}}$ using approximate values for these two lengths. Candidates who correctly found that $\dfrac{\text{ED}}{\text{CB}} = 2\cos\theta$ usually went on to obtain the correct ratio although many did not show sufficient steps of working to explain how they obtained the given answer.	
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
28			$4\sin\theta\cos\theta = 1 + 2\cos^2\theta - 1$	M1*	Use of correct double angle formulae: $\sin 2\theta \equiv 2\sin\theta\cos\theta$ **and** any one of $\cos 2\theta \equiv \cos^2\theta - \sin^2\theta$ or $1 - 2\sin^2\theta$ or $2\cos^2\theta - 1$	
			$2\cos\theta(2\sin\theta - \cos\theta) = 0$	A1	Correct equation in solvable form e.g. $2\sin\theta - \cos\theta = 0$(oe) or $5\sin^4\theta - 6\sin^2\theta + 1 = 0$ or $5\cos^4\theta - 4\cos^2\theta = 0$ but not $4\sin\theta\cos\theta = 2\cos^2\theta$	
			$\Rightarrow \tan\theta = \frac{1}{2}$	M1dep*	Use of $\dfrac{\sin\theta}{\cos\theta} \equiv \tan\theta$ on their $\alpha\sin\theta + \beta\cos\theta = 0$ or correct method for solving quadratic in either $\sin^2\theta$ or $\cos^2\theta$ (See guidance in question 2 for solving quadratics)	
			$\theta = 26.6°$	A1	www (26.6 or better)	
			$\theta = 90°$	B1	Not from incorrect working Ignore additional solutions outside the range. If any additional solutions given inside the range $0 \le \theta \le 180°$ **and** full marks would have been awarded then remove last mark (so 4/5) Both answers in radians: 0.464 (or better) and $\pi / 2$ scores B1 Answers with no working scores B1 B1(so max 2/5) **Examiner's Comments** The majority of candidates correctly replaced $\sin 2\theta$ with $2\sin\theta\cos\theta$ and $\cos 2\theta$ with one of $\cos^2\theta - \sin^2\theta$ or $1 - 2\sin^2\theta$ or $2\cos^2\theta - 1$, although a minority of	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					candidates made the costly mistake of forgetting the 2 in the latter two identities. While a majority of candidates correctly obtained $2\sin\theta\cos\theta = \cos^2\theta$ many then cancelled $\cos\theta$ from both sides of the equation instead of factorising to obtain $\cos\theta(2\sin\theta - \cos\theta) = 0$ and so lost the solution to the equation $\cos\theta = 0$. Most candidates correctly simplified $2\sin\theta - \cos\theta = 0$ to $\tan\theta = \frac{1}{2}$ and obtained the correct answer of 26.6° although a small minority gave an answer in radians or additional answers both inside and outside of the given range. A number of candidates, after applying suitable double angle identities, squared their equation leading to either a quadratic equation in either $\sin^2\theta$ or $\cos^2\theta$. These attempts usually contained sign and/or algebraic errors or even, when the correct disguised quadratic was obtained, and solved, additional incorrect solutions (coming from the earlier squaring of the single angle equation) were given.	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance	
29			$\cos\theta - 3\sin\theta = R(\cos\theta\cos\alpha - \sin\theta\sin\alpha)$ $\Rightarrow 1 = R\cos\alpha, 3 = R\sin\alpha$	M1A1	Correct pairs. Condone sign errors for the M mark (so accept $R\sin\alpha = -3$)	
			$R^2 = 1^2 + 3^2 = 10 \Rightarrow R = \sqrt{10}$	B1	Or 3.2 or better, not $\pm\sqrt{10}$ unless $+\sqrt{10}$ chosen	
			$\tan\alpha = 3 \Rightarrow \alpha = 1.249$	M1 A1	ft their pairs (condone sign errors but division must be the correct way round), A1 for 1.249 or better (accept 1.25), with no errors seen in method for angle	
			Maximum value of $\cos\theta - 3\sin\theta$ is $\sqrt{10} < 4$	B1	Or equivalent convincing numerical statement that no solutions exist e.g. $\frac{4}{\sqrt{10}} > 1$. Maybe embedded in an attempt at a solution. Do not accept general statements e.g. 'doesn't work' – must be clear why no solutions exist – **dependent on first B1** **SC**: If candidates state that $\cos\alpha = 1, \sin\alpha = 3 \Rightarrow \tan\alpha = 3$ this could score M0A0B1M1A1B1 (so max 4/6) **Note that those candidates who state $R = \sqrt{10}$ and $\tan\alpha = 3$ with no (wrong) working seen could go on to score full marks** **Examiner's Comments** The majority of candidates	

Question			Answer/Indicative content	Marks	Part marks and guidance	
					correctly worked out the values of R and α although some candidates lost the first method mark by not including R in the expanded trigonometric statements $R \cos \alpha = 1$ and $R \sin \alpha = 3$. Some failed to give α in radians and a small minority stated R as 10 rather than the correct $\sqrt{10}$. Candidates were less successfully in showing that $\cos \theta - 3 \sin \theta = 4$ had no solutions with many simply $\theta + 1.249 = \arccos\left(\frac{4}{\sqrt{10}}\right)$ 'does not work' or gives a 'math error'. Many candidates failed to explain or give an equivalent mathematical statement that the maximum value of $\cos \theta - 3 \sin \theta$ is $\sqrt{10}$ which is less than 4 and so did not score the final mark in this question.	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
30		i	$\alpha = -23.44 \times \cos\left(\frac{360}{365} \times (n+10)\right)$			
		i	On February 2nd, $n = 31 + 2 = 33$	B1	calculate $n = 31 + 2 = 33$ (days in Jan + Feb) **soi**	SC B1 condone $30 + 2$
		i	$\alpha = -23.44 \times \cos\left(\frac{360}{365} \times 43\right)$	M1	substitution of their $n + 10$ in equation (3) and attempt to evaluate	Where $n = 31, 32, 33, 34$ only
		i	$a = -17.31$	A1	or -17.306 or rounds to -17.3 **Examiner's Comments** Most candidates found ? = -17.31 as required. A few chose to use the number of days in January as 30 and lost one mark. Those who thought February was the first or third month received no credit.	NB $n = 32 + 10$ gives -17.576 gaining B1M1A0
		ii	$\tan y = -\frac{1}{\tan \alpha}\cos(15t)$			
		ii	$\tan 53 = -\frac{1}{\tan(-17.306)} \times \cos(15t)$	M1	use of **their** α in equation (4)	
		ii	$t = \frac{1}{15}\arccos(-\tan 53 \times \tan(-17.306))$	DM1	making t the subject	
		ii	$t = 4.3717$	A1	cao	**SC ft** from -17.576 Obtains A1ft for 4.343 (or 4.34)
		ii	Sunset is at 12 hrs + 4 hours 22 minutes, and so 16:22 hrs	A1		And then A1ft for 16:21 (or 16:20)

Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii			<u>Examiner's Comments</u> Some candidates carelessly lost the negative sign in their angle or the negative sign in the formula and so lost unnecessary marks. Many correctly obtained t =4.37 but not all converted this correctly to the 24 hour clock. 16:37 was commonly seen. Candidates were able to follow through for full marks from the Special Case in part(ii)	
			Total	7		

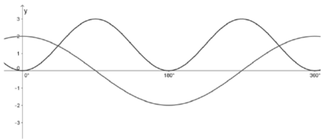
Question			Answer/Indicative content	Marks	Part marks and guidance	
31			$\tan y = -\frac{1}{\tan\alpha}\cos(15t)$			
			? = 23.44°, y = 60°, t is to be found			
			$\cos(15t) = -0.7509\ldots$ $15t = 138.6737\ldots$	M1	substitute in formula and attempt to solve (as far as $15t$ = invcos … …) oe	
			$t = 9.2449\ldots$	A1	accept 9.2 or better	
			Daylight hours are 2 × 9.2449… = 18.4898 …	DM1	doubling, dependent on first M1	
			So 18.5 hours (to 3s.f.)	A1	or approx 18 hours www (accept 18.4898, 18.489, 18.49, 18.5 or 18.4 (from 2 × 9.2), 18.48)	any reasonable accuracy or stating error is approx 0.49 oe
			OR			
			Using t = 9 ? = 23.44, t = 9, y is to be found	M1	t = 18/2 = 9 and substituted in formula	any reasonable accuracy
			$\tan y = 1.6309\ldots\ldots$	DM1	and attempt to solve (as far as y = inv tan constant) oe	
			y = 58.485° so approx 60°	A2	or 58.49° / 58.5° / approx 60° www __Examiner's Comments__ Most candidates substituted α =60° and found t=9.2449 or similar and then the majority multiplied it by 2 to compare it with18. A few worked backwards from t=9 to reach approximately 60° when substituting in the appropriate equation, and were given full credit.	
			Total	4		

Question			Answer/Indicative content	Marks	Part marks and guidance	
32			10° north is B 1° north is A 5° south is D 15° south is C	B2	All four answers correct **SC B1** Any two answers correct Examiner's Comments The graphs were usually identified correctly although there were also many guesses. The response tended to be either fully correct or all wrong.	
			Total	2		
33			For 30° S, 10 hours daylight	B1	cao soi	10 only
			For 60° S, 5.5 hours daylight.	B1	allow $5 < t < 6$ soi	not 5 or 6
			Difference 4.5 hours	B1ft	10-t dependent on both previous B marks soi Examiner's Comments Most candidates had the right idea but some were inaccurate with 10, 6, 4 being the most common alternative solution.	**SC(1)** allow B3 for $4 <$ difference in length < 5 (**without wrong working**) **SC(2)** allow B2 if uses $t = 5$ or $t = 6$ eg, 10,6,4 www **SC(3)** If calculates (using the equation (4)) can obtain all three marks. Approximate values are 10.07, 5.51, 4.56. (May also work with 5 and $2.5 < t < 3$ if doubled at end, e.g. 5 B1 dependent on later doubling 2.6 B1 dependent on later doubling $(5 – 2.6) \times 2 = 4.8$ B1)
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
34			Point R marked correctly at $(-60, 45)$ Point M marked correctly at $(0, 20)$	B1 B1	half way up the relevant square Generously applied if above half way If unclear, B0B0. **Examiner's Comments** Answers were often correct but surprisingly many, and various, incorrect positions were seen. A number of candidates only used the letter *R* or *M* to indicate their points where the addition of a cross or dot would have made their position clearer.	Need labelling with letters or co-ordinates oe (condone one not labelled if other is labelled (and no others marked)) **SC B1** BOD both marked (and no others) and no labelling.
			Total	2		
35			$\int_0^{\frac{\pi}{12}} \cos 3x\,dx = \left[\frac{\sin 3x}{3}\right]_0^{\frac{\pi}{12}}$ $= \frac{1}{3}\left(\sin\frac{\pi}{4} - 0\right)$ $= \frac{\sqrt{2}}{6}$ o.e.	B1(AO1.1) M1(AO1.1) A1(AO1.1) [3]	$\frac{\sin 3x}{3}$ Must be in exact form	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance
36		a	$\cos A = \frac{100^2 + 120^2 - 135^2}{2 \times 100 \times 120}$	M1(AO3.1a)	cos B = $\frac{100^2 + 135^2 - 120^2}{2 \times 100 \times 135}$ OR cos C = $\frac{120^2 + 135^2 - 100^2}{2 \times 120 \times 135}$
			cos A = 0.2572916…..	A1(AO1.1)	cos B = 0.512037….. OR cos C = 0.698302….. (may be implied)
			[A =] 75.09058…	A1(AO1.1)	B = 59.200… OR C = 45.7090…
			Area = $\frac{1}{2}$ ×100×120× sin(*their A*)	M1(AO3.1a)	Area = $\frac{1}{2}$ ×100×135× sin(*their B*) OR $\frac{1}{2}$ ×100×135× sin(*their C*)
			5800 [m²]	A1(AO1.1)	Accept answers to greater degree of accuracy
				[5]	

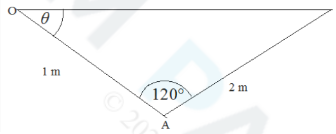
Question			Answer/Indicative content	Marks	Part marks and guidance	
		b	E.g. The sides might only be to the nearest 5 metres so the possible areas cover quite a big range E.g. The sides are no more accurate than to the nearest metre, so could be half a metre out. Taking half a metre off each side would lose more than 1 m^2 of area	E1(AO3.2b) [1]	Correct explanation	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance	
37		a		B1(AO1.1a) B1(AO1.1) [2]	Correct shape and symmetry for cosine graph. Correct maximum and minimum values	
		b	DR $2\cos\theta = 3\sin^2\theta$ $2\cos\theta = 3(1-\cos^2\theta)$ $3\cos^2\theta + 2\cos\theta - 3 = 0$ $\cos\theta = \frac{-1}{3} + \frac{\sqrt{10}}{3}$ $\theta = 43.9°, 316.1°$ $\cos\theta = \frac{-1}{3} - \frac{\sqrt{10}}{3} < -1$ gives no solution	B1(AO1.2) M1(AO3.1a) M1(AO1.1) A1(AO1.1) A1(AO1.1) E1(AO2.4) [6]	Correct use of identity **must be seen** Rearranging to zero **must be** seen, condone one error Solve quadratic Or state that graph in part (a) only shows two solutions	

Question			Answer/Indicative content	Marks	Part marks and guidance	
			Total	8		
38			$\arcsin x = \theta$ $\Rightarrow x = \sin\theta$ $\arccos y = \theta \Rightarrow y = \cos\theta$ $\sin^2\theta + \cos^2\theta = 1$ $\Rightarrow x^2 + y^2 = 1$ AG	M1(AO1.1) M1(AO1.1) E1(AO2.1) [3]		
			Total	3		
39			Area = $5 = \frac{1}{2} \times 7^2 \times \theta$ $\theta = \frac{10}{49} [= 0.204]$	M1(AO3.1a) A1(AO1.1) [2]		
			Total	2		
40		a	Horizontal component = 100 × cos32° 84.8 N	M1(AO3.3) A1(AO1.1) [2]		
		b	Frictional force is 100 cos θ [with $\theta < 32°$] So frictional force increases. (oe)	M1(AO3.4) E1(AO2.2a) [2]		
			Total	4		

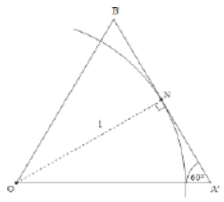
Question			Answer/Indicative content	Marks	Part marks and guidance		
41			DR Radius = $\sqrt{10}$ $\cos C = \dfrac{10+10-(7-1)^2}{2\times\sqrt{10}\times\sqrt{10}}$	B1(AO1.1) M1(AO3.1a)	Or use right angled triangle: $\cos x = \frac{3}{\sqrt{10}}$ and M1 for $\frac{1}{2}C = \frac{\pi}{2} - \cos^{-1}\left(\frac{3}{\sqrt{10}}\right)$		
			C = 2.50 (3sf) $\text{Area} = \frac{1}{2}\times(\sqrt{10})^2\times 2.50 - \frac{1}{2}\times(\sqrt{10})^2\times\sin 2.50$ Area = 9.49	A1(AO1.1) M1(AO3.1a) A1(AO1.1) [5]			
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance	
42		a	BAC = 360 – 120 – 90 – (90 – θ) = θ + 60 ⇒ BC = 2 sin(θ + 60) CD = AE = sin θ ⇒ h = CD + BC = sin θ +2 sin (θ + 60°)	B1(AO3.1a) M1(AO1.1) E1(AO2.1) [3]	AG	

Question			Answer/Indicative content	Marks	Part marks and guidance
		b	$h = \sin\theta + 2\sin(\theta + 60°)$ $= \sin\theta + 2(\sin\theta\cos 60 + \cos\theta\sin 60)$	M1(AO3.1a)	use of compound angle formula
			$= \sin\theta + \sin\theta + \sqrt{3}\cos\theta$	A1(AO2.1)	
			$= 2\sin\theta + \sqrt{3}\cos\theta$	M1(AO1.1)	
			$h = 0$ $2\sin\theta + \sqrt{3}\cos\theta = 0$		$h = 0$ soi
			$\Rightarrow$	M1(AO1.1)	Use of
			$\Rightarrow \tan\theta = -\frac{\sqrt{3}}{2}$	A1(AO1.1)	$\frac{\sin}{\cos} = \tan$
			$\Rightarrow \theta = -40.9°$ [so 40.9° below the horizontal]	M1(AO3.1a)	or 319.1° or 139.1°
			Alternative method Diagram with $h = 0$ 	M1(AO2.1)	
			$a^2 = 1^2 + 2^2 - 4\cos 120°$	A1(AO1.1)	
			$a = \sqrt{7}$	M1(AO1.1)	
			$\sin\theta = \frac{2\sin 120°}{\sqrt{7}} = \sqrt{\frac{3}{7}}$	A1(AO1.1)	
			$\theta = -40.9°$[so 40.9° below the horizontal]	[5]	For final mark, θ shown below horizontal in diagram together with 40.9° is accepteble
			Total	8	

Question			Answer/Indicative content	Marks	Part marks and guidance	
43		a	cos θ + 2sin θ ° R cos(θ – α) ⇒ Rcos α = 1, Rsin α = 2 ⇒ R2 = 5, R = $\sqrt{5}$ tan α = 2, α = 1.107	M1(AO1.1a) B1(AO1.1) M1A1(AO1.1 1.1) [4]		
		b	max value is $\frac{1}{\left(k - \sqrt{5}\right)}$ $\frac{1}{\left(k - \sqrt{5}\right)} = \frac{\left(3+\sqrt{5}\right)}{4}$ $4 = 3k - 5 + k\sqrt{5} - 3\sqrt{5}$ [This is indep of $\sqrt{5}$ so] k = 3	M1(AO3.1a) M1(AO1.1) A1(AO1.1) [3]		
			Total	7		

Question			Answer/Indicative content	Marks	Part marks and guidance	
44		a	(i) $f'(x) = e^{-x}\cos x - e^{-x}\sin x$ $f'(x) = 0$ and $e^{-x} \neq 0 \Rightarrow \cos x = \sin x$ $\Rightarrow \tan x = 1$ $\Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \frac{13\pi}{4}$	M1(AO3.1a) A1(AO1.1) E1(AO2.2a) M1(AO1.1) A1(AO1.1)	product rule correct Use of $\frac{\sin}{\cos} = \tan$ $x = \frac{\pi}{4}$ (condone 45°) $\frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, \ldots$	
		a	So an AP with $d = \pi$ (ii) $y = \frac{\sqrt{2}}{2}e^{-\frac{\pi}{4}}, -\frac{\sqrt{2}}{2}e^{-\frac{5\pi}{4}}, \frac{\sqrt{2}}{2}e^{-\frac{9\pi}{4}}, -\frac{\sqrt{2}}{2}e^{-\frac{13\pi}{4}}$ This is a GP with $r = -e^{-\pi}$	E1FT(AO2.1) M1(AO3.10a) A1(AO1.1) E1FT(AO2.1) [9]	must state the common difference substituting one value of x into $f(x)$ must state common ratio, www	FT their values of x FT their values of y
		b	Yes with explanation that values of x would continue to be separated by pi and so values of y would continue to have same common ratio.	E1(AO2.2a) [1]		
			Total	10		

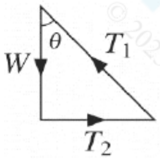
Question			Answer/Indicative content	Marks	Part marks and guidance	
45			Each triangle (like OAB) is equilateral	E1(AO2.1) [1]	Oe	
			Total	1		
46			 Show diagram which was previously fig 13 Angle A' N = tan 30° OR $\tan 30^\circ = \frac{1}{A'N}$ $A'N = \tan 30^\circ = \frac{1}{\sqrt{3}}$ **Alternative method** using the equilateral triangle OA' B' of side length $2a$: $(2a)^2 = a^2 + 1 \Rightarrow a^2 = \frac{1}{3}$ $a = A'N = \frac{1}{\sqrt{3}}$ Evidence of 6 × A'B' or 12 × A'N = $4\sqrt{3}$	M1(AO3.1a) A1(AO1.1) M1(AO3.1a) M1(AO1.1) E1(AO2.4) [3]	Soi AG	
			Total	3		

Question			Answer/Indicative content	Marks	Part marks and guidance	
47		a	Angle = 360 ÷ 24 = 15 Edge length = 2 tan15° Perimeter = 12 × 2 tan15° = 24 tan15°	M1(AO1.1) E1(AO2.1) [2]	AG	
		b	tan15° = tan (45° – 30°) $=\frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}}$ $\left[=\frac{\sqrt{3}-1}{\sqrt{3}+1}=\frac{\left(\sqrt{3}-1\right)^2}{2}\right]$ **Alternative method** tan15° = tan (60° – 45°) $=\frac{\sqrt{3}-1}{1+\sqrt{3}}$ $\left[=\frac{2\sqrt{3}-4}{-2}\right]$ Perimeter = 12 × 2 tan15° $=48-24\sqrt{3}$	B1(AO3.1a) M1(AO1.1) B1(AO3.1a) M1(AO1.1) E1(AO2.1) [3]	Exact values of tan 45° and tan30° used Exact values of tan 60° and tan15° used Correct completion AG	
			Total	5		

Question			Answer/Indicative content	Marks	Part marks and guidance		
48			$\frac{2\sin\theta}{\cos\theta}+\cos\theta=0$ $2\sin\theta + 1 - \sin^2\theta = 0$ $\sin\theta = 1 \pm \sqrt{2}$ $\sin\theta = 1+\sqrt{2}$ has no roots since $-1 \le \sin\theta \le 1$ If $\sin\theta = 1-\sqrt{2}$, $\theta = -24.47$ or $-155\text{-}53$ 204 335	M1(AO1.1) M1(AO3.1a) A1(AO1.1) E1(AO2.3) A1(AO1.1) A1(AO3.2a) A1(AO1.1) [7]	DR Use of identity Multiplication by cos θ and use of Pythagoras Both answers from correct factorizing or correct use of quadratic formula allow 204.5 or 204.47 allow 335.5 or 335.53	Ignore extra values outside range. Deduct one mark if extra values in range. If A0A0 allow SC1 for both correct answers given to greater precision.	
			Total	7			

Question			Answer/Indicative content	Marks	Part marks and guidance		
49			$2\theta = \cos^{-1} 0.3$ $2\theta = 72.54, 287.46, 432.54, 647.46$ $\theta = 36.3, 143.7, 216.3, 323.7$	M1(AO 1.1a) A1(AO 1.1b) A1(AO 1.1b) [3]	 For at least one correct answer in the range For all correct with no others in the range		
			Total	3			
50			$v = \frac{ds}{dt} = 3 + 0.3t^2$ and $a = \frac{dv}{dt} = 0.3 \times 2t$ $a = 0.6t$ Acceleration is a function of t so is not constant **Alternative method** For constant acceleration, $s = ut + \frac{1}{2}at^2\ (+c)$ i.e. s is a quadratic function of t So this cubic function is not constant acceleration	M1(AO 1.1a) A1(AO 2.1) E1(AO 2.2a) M1 A1 E1 [3]	Attempt to differentiate twice Must be clearly argued Comparing with standard formula Explicit identification of quadratic, oe Deduction clearly stated		
			Total	3			

Question			Answer/Indicative content	Marks	Part marks and guidance	
51		a	Area of triangle $=\frac{1}{2}\times 10^2 \sin\frac{1}{6}\pi \quad (=25)$ $AC = 2\times 10\sin\frac{1}{12}\pi \quad (=5.17638)$ $=\frac{1}{2}\pi\times\left(\frac{5.17638}{2}\right)^2 \quad (=10.5223)$ Area of semicircle Total area = 35.5 cm^2	B1(AO 1.1a) M1(AO 3.1a) M1(AO 1.1b) A1(AO 1.1a) [4]	Correct unsimplified expression soi Or use of cosine rule Use of $\frac{1}{2}\pi\times$(half of their AC)^2	
		b	$50\theta = 35.52$ $\theta = 0.710$ to 3sf	M1(AO 1.1a) A1(AO 1.1b) [2]	Equating $\frac{1}{2}r^2\theta$ to their answer from **(a)**	
			Total	6		

Question			Answer/Indicative content	Marks	Part marks and guidance
52			T_1 and T_2 are tensions in wire and string respectively; the globe has mass m		
			Resolve vertically: $T_1 \cos\theta = mg$	M1(AO 3.1b)	Allow sin/cos interchange for 1st M1 only
			$T_1 = \dfrac{mg}{\cos\theta}$	M1(AO 3.3)	
			Resolve horizontally: $T_2 = T_1 \sin\theta$	M1(AO 2.1)	Use of trig identity
			$T_2 = \dfrac{mg}{\cos\theta} \times \sin\theta = mg\tan\theta$		
			$mg\tan\theta < mg$ for $\tan\theta < 1$	A1(AO 2.2a)	
			$\theta < 45°$		Inequality clearly stated
			Alternative method	M1	
			 Triangle of forces	M1	Forces shown, correct arrow directions
			Maximum T_2 is when $T_2 = W$	M1	Equating T_2 and weight for limiting case
			In this case $\theta = \arctan(1)$	A1	Use of trigonometry or isosceles triangle
			Hence $\theta < 45°$	[4]	Inequality clearly stated
			Total	4	

Question			Answer/Indicative content	Marks	Part marks and guidance	
53		a	$\sec\theta - \cos\theta \equiv \frac{1}{\cos\theta} - \cos\theta$ $\equiv \frac{1-\cos^2\theta}{\cos\theta}$ $\equiv \frac{\sin^2\theta}{\cos\theta} \equiv \frac{\sin\theta}{\cos\theta}\sin\theta$ $\equiv \tan\theta\sin\theta$ AG	B1(AO 1.2) M1(AO 2.1) M1(AO 1.2) E1(AO 2.1) [4]	Use of definition of sec Manipulation of fractions Use of at least one trig identity Clear and complete argument	

Question			Answer/Indicative content	Marks	Part marks and guidance		
		b	$\tan\theta\sin\theta = \frac{1}{2}\tan\theta \Rightarrow \tan\theta\left(\sin\theta - \frac{1}{2}\right) = 0$ $\tan\theta = 0$ gives $\theta = 0, \pi$ $\sin\theta = \frac{1}{2}$ gives $\theta = \frac{1}{6}\pi, \frac{5}{6}\pi$ Alternative method $1 - \cos^2\theta = \frac{1}{2}\sin\theta \Rightarrow \sin^2\theta = \frac{1}{2}\sin\theta$ $\sin\theta = 0$, gives $\theta = 0, \pi$ $\sin\theta = \frac{1}{2}$ gives $\theta = \frac{1}{6}\pi, \frac{5}{6}\pi$	M1(AO 1.1a) A1(AO 1.1b) M1(AO 1.1a) A1(AO 1.1b) M1 A1 M1 A1 [4]	Factorising Both values At least one value from arcsin Both values correct Obtaining quadracit in $\sin\theta$ Both values At least one value from arcsin Both values correct	SCI for sole answer $\frac{1}{6}\pi$ www	
			Total	8			

Question			Answer/Indicative content	Marks	Part marks and guidance	
54		a	$a = 40$ $c = 180$ $30 = 40 - b(315 - 180)^2$ $\Rightarrow b = \frac{2}{405}$ **AG**	B1(AO 3.3) B1(AO 3.3) B1(AO 2.1) [3]	At least one intermediate step needed	
		b	(*A*) 15.8	B1(AO 3.4) [1]	Allow integer answers 15 or 16	
		b	(*B*) 250 is outside the interval which provides the data and so may be extrapolating too far	E1(AO 3.5a) [1]		
		c	(*A*) $t < 90 \Rightarrow N < 0$ which is not meaningful	E1(AO 2.4a) [1]		
		c	(*B*) $t > 270$	E1(AO 3.5b) [1]		
		d	Reflection in t-axis or translated by $\begin{pmatrix}180\\0\end{pmatrix}$ or $\begin{pmatrix}-180\\0\end{pmatrix}$ Stretch in the y-direction, scale factor 40	B1(AO 1.1a) B1(AO 1.1a) [2]	Vector notation required for translation	Accept stretch in the y -direction, scale factor –40, for **B2**
		e	$N = -40\cos t$ or $N = 40\cos(t - 180)$	B1(AO 3.3) [1]	oe, e.g. $N = 40\cos(t + 180)$	

Question			Answer/Indicative content	Marks	Part marks and guidance	
		f	(*A*) Stretch graph in *t* -direction by scale factor $\frac{365}{360}$ or modify equation to $N = -40\cos\left(\frac{360}{365}t\right)$	B1(AO 3.5c) [1]	oe, e.g. $N = 40\cos\left(\frac{360}{365}t - 180\right)$; FT from their (**v**) for modified equations	
		f	(*B*) Maximum point moves to (182.5, 40) **or** as consistent with their answer to (*A*)	B1(AO 3.4) [1]	N.B: $N = 40\cos\left(\frac{360}{365}(t-180)\right)$ oe will 365 leave maximum point unchanged	
			Total	12		

Question			Answer/Indicative content	Marks	Part marks and guidance	
55		a	$2\cos\theta + 3\sin\theta \equiv R\sin(\theta + \alpha)$ $\Rightarrow R\cos\alpha = 3,\ R\sin\alpha = 2$	M1(AO 1.1a)		
			So $R^2 = 13 \Rightarrow R = \sqrt{13}$	B1(AO 1.1)		
			and $\tan\alpha = \frac{2}{3}$	M1(AO 1.1)		
			$\Rightarrow \alpha = 0.588$	A1(AO 1.1)		
				[4]		
		b	$k + 2\cos x + 3\sin x > 0$ [for all x]	B1(AO 3.1a)	oe	
			$\sqrt{13}\sin(x + 0.588) + k > 0$	M1(AO 1.1)	Use of expression from part **(a)**	
			Minimum value of LHS is $k - \sqrt{13}$	M1(AO 3.1a)	Attempt to find minimum value	May be by calculus
			$k > \sqrt{13}$	A1(AO 2.2a)		
				[4]		
		c	$k + 2\cos x + 3\sin x > 0 \Rightarrow \frac{1}{k + 2\cos x + 3\sin x} > 0$	E1(AO 2.4)	oe; accept e.g. statement that the reciprocal of a positive number is positive	
				[1]		

Question			Answer/Indicative content	Marks	Part marks and guidance		
			Total	9			
56		a	$\tan 45^0 = 1$ and $\tan 60^0 = \sqrt{3}$	B1(AO 1.2) [1]			
		b	$\tan(60-45) = \dfrac{\tan 60 - \tan 45}{1 + \tan 60 \times \tan 45}$ $\dfrac{\sqrt{3}-1}{1+\sqrt{3}}$ Multiply numerator and denominator by $\sqrt{3}-1$ eg $\dfrac{3-2\sqrt{3}+1}{3-1}$ $= 2 - \sqrt{3}$	M1(AO 3.1a) M1(AO 1.1) M1(AO 1.1) A1(AO 2.1) [4]	DR Substitution of their surds in correct compound angle formula AG Convincing arithmetic to given result	Other correct methods eg use of double angle formula are acceptable	
			Total	5			

Question			Answer/Indicative content	Marks	Part marks and guidance		
57			Use of cosine rule $\cos\theta = \frac{47^2 + 53^2 - 94^2}{2 \times 47 \times 53}$ 140° or 2.44 radians	M1(AO 3.1) A1(AO 1.1) A1(AO 1.1) [3]	allow if lengths incorrectly placed NB – 0.766 35889… allow 140.0 ,140.03 or 140.028 or 2.444, 2.4440 or 2.44395	NB 21.23606 or 0.370(6393 …); or 18.73589 or 0.327(0030 …)	
			Total	3			